Thermal dimension of the one-dimensional long-range Ising model from bosonic NRG of the sub-ohmic spin–boson model

Method, data, checks, results, limitations and reproduction

Z. Zheng

Status of 15 October 2026

Abstract

We compute the scaling dimension Δ_ε of the energy operator of the one-dimensional long-range Ising model, with couplings decaying as $1/r^{1+s}$, at thirteen values of the spin dimension $\Delta_\sigma = (1-s)/2$ between 0.249 and 0.001. The critical point is realized as the quantum critical point of the sub-ohmic spin–boson model, which we solve with a bosonic numerical renormalization group (NRG); the relevant eigenvalue $y_t = 1 - \Delta_\varepsilon$ is obtained from the exact derivative of the truncated NRG map with respect to the bath coupling. At ten points, $0.001 \le \Delta_\sigma \le 0.2$, we give final values, for example $\Delta_\varepsilon = 0.529608(33)$ at $s = 0.8$ and 0.4830(7) at $s = 0.6$. At eight of them, $0.001 \le \Delta_\sigma \le 0.15$, the local exponent settles on a plateau along the chain, and the error scales are (2.7–5.4) $\times 10^{-5}$ for $0.01 \le \Delta_\sigma \le 0.125$, 1.1×10^{-4} at $\Delta_\sigma = 0.15$ and 1.1×10^{-4} at $\Delta_\sigma = 0.001$. At $\Delta_\sigma = 0.175$ and 0.2 the local exponent is still drifting at the end of the usable chain and is extrapolated, with error scales 3.5×10^{-4} and 7.1×10^{-4}. Each error adds three parts linearly: the drift along the chain, the spread over kept states and oscillator levels, and a discretization error, the larger of the spread over the values of Λ that were run and the change of the $\Lambda = 2.5$ value when the number of kept states is doubled; the latter sets it at 8 of the 10 final points. The errors are not standard deviations. Near $s = 1/2$ the leading correction exponent $\omega \approx \varepsilon$ vanishes, and within the chain depth reachable in double precision the flow does not reach its fixed point: at $\Delta_\sigma = 0.249$, 0.24 and 0.225 the values where it comes closest lie $+8.7 \times 10^{-3}$, $+4.7 \times 10^{-3}$ and $+1.6 \times 10^{-3}$ above the ε expansion, and at $\Delta_\sigma = 0.249$ the inverse excess grows along the renormalization-group time at a rate of the size that one loop predicts. We label these three points not converged in chain depth, a label assigned after the comparison with the series, show their values, and use them in no comparison. Near $s = 1$ the results reproduce the strong–weak dual expansion of Δ_ε through its fifth coefficient. At the 5 points where the project's cluster Monte Carlo study accepts its own value, NRG differs from it by at most 2.9×10^{-3}. That is within the sum of the Monte Carlo statistical error, its fit-model half-range and the NRG error at 4 points, and 1.08 times that sum at the fifth; but it is up to 14 Monte Carlo statistical errors, with the same sign at neighboring points, so only the fit-model half-range reconciles the two. The Monte Carlo crossing extrapolations agree with NRG within 1.05 of their statistical errors at these points. A second solver, written by a separate Claude agent from a physics-only specification, reproduces the chain coefficients to better than 10^{-15} (rounding level), the critical couplings bitwise or to 7.2×10^{-14}, and Δ_E to 3.1×10^{-7} in every window its finite differences can reach. The code was written by GPT-6 astra (OpenAI) and ported to the cluster, run and analyzed by Claude Code (Anthropic); every number in this note is regenerated from the released records by `make replay`.

Contents

1 Overview **3**
1.1 Scope . . . 3
1.2 Headline results . . . 3

2 The problem and what is known **4**
2.1 The long-range Ising chain and its thermal dimension . . . 4
2.2 The spin–boson model . . . 5
2.3 The thirteen points . . . 6

3 Method: bosonic NRG with an exact α-derivative **6**
3.1 Logarithmic discretization . . . 6
3.2 Wilson chain . . . 6
3.3 Iterative diagonalization and truncation . . . 6
3.4 Parity-gap flow and critical coupling . . . 7

3.5 Exact α-derivative of the truncated map . . . 7
3.6 The window audit . . . 8
3.7 The estimate: settled value, plateau and extrapolation . . . 8
3.8 Cross-check fits (not adopted) . . . 9
3.9 Chain depth . . . 9

4 Error budget **9**

5 Software and validation **10**
5.1 Code . . . 10
5.2 Acceptance gates and tests . . . 11
5.3 Reproducibility across machines . . . 12

6 Production campaigns **12**

7 Results **12**
7.1 The approach to the fixed point . . . 12
7.2 Convergence in kept states and oscillator levels . . . 13
7.3 Dependence on the discretization Λ . . . 14
7.4 The final table . . . 15
7.5 The limit $s \to 1$ . . . 15

8 Checks **16**
8.1 A second implementation . . . 16
8.2 The ε expansion near $s = 1/2$ . . . 18
8.3 The 1D cluster Monte Carlo study . . . 20
8.4 Published values . . . 21

9 Limitations and open problems **21**

10 Reproduction **22**
10.1 The released folder . . . 22
10.2 How to run . . . 22

A Data formats **24**

B Production on the cluster **24**

C Provenance and review **25**

D All production items **27**

E The superseded midpoint estimator **29**

F Terms and symbols **29**

1 Overview

1.1 Scope

This note is written for a physicist outside the project. Section 2 states the problem and what is known analytically. Sections 3 and 4 describe the numerical method, the estimator and the error budget in enough detail to re-implement them. Section 5 describes the code and how it was validated, Sec. 6 the production runs, Sec. 7 the results, and Sec. 8 the checks against a second implementation, against two methods that do not use NRG and against published values. Section 9 collects the limitations and Sec. 10 explains how to recompute every number. The appendices give the data formats (App. A), the production on the cluster (App. B), the provenance of code, analysis and review (App. C), every production run (App. D), the estimator used before this revision and its values (App. E), and the terms and symbols (App. F).

Who did what. The NRG solver was written by GPT-6 astra (OpenAI, through ChatGPT) in two stages: a reference solver for an earlier, unreleased study of the project (the *original study*; shipped as `code/lri_spin_boson_hpc/reference/`) and an optimized package built on it. Claude Code (Anthropic) reviewed the package, ported it to a Slurm cluster, ran the production, wrote the analysis and drafted this note; OpenAI Codex reviewed the cluster port. Two rounds of read-only reviews of the analysis and of this note, three reviews each, were made by Claude agents on 13 October 2026, and an independent reader reviewed the text on 15 October. This version follows them; the changes made after the second round and after the reading were not reviewed again. There was no human referee. App. C gives the details, including which model ran where. In this note "the author" is Z. Zheng, who made the decisions marked as his, and "we" is the study as a whole: the author and the AI systems that worked for him.

Vocabulary. A *point* is one value of Δ_σ. An *item* is one complete NRG calculation at fixed $(\Delta_\sigma, \Lambda, K, n_b, N)$: the search for the critical coupling, the exact derivative of the critical flow, and the window audit (Sec. 3). Its *record* is the JSON file that holds the result. The *central item* of a point is the one at $\Lambda = 3$, $K = 128$, $n_b = 16$; all other items are convergence scans. A *campaign* is a set of items submitted together (Sec. 6), and *production* means the 197 items whose records are released. A *window* is a stretch of consecutive chain sites over which the growth of the derivative is fitted (Sec. 3.6). We write Δ_E for the NRG estimate of Δ_ε; despite the notation it is not an energy difference. App. F lists all terms and symbols.

1.2 Headline results

Table 1 is the main result, and Fig. 1 shows it with the references of Sec. 8. "Final" values are the estimates of Sec. 3.7 with the total error $\delta_{\rm tot}$ of Sec. 4. The three values in gray brackets are not converged in chain depth (Sec. 7.1): they are shown, carry no error, and enter no comparison and no agreement statement in this note. That label is not an outcome of the estimator; it was assigned after the comparison with the ε series (Sec. 3.7).

What the checks establish:

- **The code computes what it claims.** A second implementation written from a physics-only specification (Sec. 8.1) agrees with the production solver in eight configurations: chain coefficients to better than 10^{-15} (relative, rounding level), critical couplings bitwise in 5 of 8 configurations and to 7.2×10^{-14} in the others, and Δ_E window by window to 3.1×10^{-7} wherever its finite differences are reliable. This tests the numerical map, not the estimator.
- **Comparison with methods that do not use NRG.** Near $s = 1$ the final values reproduce the dual expansion of Δ_ε in $\sqrt{\Delta_\sigma}$ through its fifth coefficient (Sec. 7.5). At the 5 points where the Monte Carlo study accepts its own value they differ from it by at most 2.9×10^{-3}: within $\sigma_{\rm stat} + \sigma_{\rm sys} + \delta_{\rm tot}$ at 4 points and 1.08 times that sum at $\Delta_\sigma = 0.1$. The sum adds a standard deviation, a fit-model half-range and an error scale and is only a scale, not a test of statistical consistency. The differences are up to 14 Monte Carlo statistical errors and have the same sign at neighboring points, so only the Monte Carlo half-range $\sigma_{\rm sys}$ reconciles the two, and the comparison tests the Monte Carlo systematics more than NRG (Sec. 8.3). At $\Delta_\sigma = 0.2$ the resummed series has a spread of 9×10^{-3} and does not test NRG.
- **Near $s = 1/2$ the chain is too short.** The flow approaches its fixed point at a rate $\omega \approx \varepsilon \to 0$ in the RG time l. At $\Delta_\sigma = 0.249$ the inverse distance to the series value grows nearly linearly in l, with slopes 2.55 to 3.03, of the size of the one-loop prediction (3.14–3.17 to 3.44–3.47; Fig. 3c, Sec. 7.1). Convergence scans in K, n_b or Λ at fixed depth cannot reveal this, because once K and $n_b \geq 12$ are converged the effect depends only on l.

Table 1: Thermal dimension Δ_ε of the $d = 1$ long-range Ising model from bosonic NRG: $\Delta_E = 1 - y_t$ at $\Lambda = 3$, $K = 128$ kept states per parity sector and $n_b = 16$ oscillator levels, with the total error $\delta_{\rm tot}$ in parentheses (Sec. 4: chain depth plus K/n_b spread plus discretization error, the last mostly an allowance for the convergence in K at $\Lambda = 2.5$; PDG rounding). "Final" is the plateau value of Sec. 3.7; "final, extrapolated" is the extrapolation of the local exponent. Gray: not converged in chain depth; the settled value, no error, not used in any comparison, and its excess over the ε series in the last column. External test: the ε series (Sec. 8.2), whose spread at $\Delta_\sigma = 0.2$ is far larger than the NRG error; the label that the Monte Carlo study gives its own value (Sec. 8.3) and, where that value is accepted, NRG minus Monte Carlo with its ratio to $\sigma_{\rm stat} + \sigma_{\rm sys} + \delta_{\rm tot}$; at $\Delta_\sigma \le 0.01$ only the expansion about $s = 1$ of Sec. 7.5.

Δ_σ	s	ε	Δ_E	status	external test (Sec. 8)
0.249	0.502	0.004	[0.5080]	not converged	series 0.499344 (excess $+8.7 \times 10^{-3}$)
0.24	0.52	0.04	[0.4989]	not converged	series 0.49422 (excess $+4.7 \times 10^{-3}$)
0.225	0.55	0.1	[0.4897]	not converged	series 0.48808 (excess $+1.6 \times 10^{-3}$); MC label: conditional
0.2	0.6	0.2	0.4830(7)	final, extrapolated	series 0.48292 (spread 9×10^{-3}: no test)
0.175	0.65	0.3	0.48390(35)	final, extrapolated	MC label: cross-check failed (not used)
0.15	0.7	0.4	0.49111(11)	final	MC label: cross-check failed (not used)
0.125	0.75	0.5	0.50567(5)	final	MC label: accepted; NRG−MC -1.4×10^{-3} (ratio 0.61)
0.1	0.8	0.6	0.529608(33)	final	MC label: accepted; NRG−MC -2.9×10^{-3} (ratio 1.08)
0.075	0.85	0.7	0.566260(30)	final	MC label: accepted; NRG−MC -6.8×10^{-4} (ratio 0.38)
0.05	0.9	0.8	0.622115(27)	final	MC label: accepted; NRG−MC $+2.2 \times 10^{-3}$ (ratio 0.81)
0.025	0.95	0.9	0.713422(29)	final	MC label: accepted; NRG−MC $+2.3 \times 10^{-3}$ (ratio 0.22)
0.01	0.98	0.96	0.80975(5)	final	$s \to 1$ expansion only
0.001	0.998	0.996	0.93742(11)	final	$s \to 1$ expansion only

The errors are error scales built from measured spreads and a few stated judgements (Sec. 4); they are not standard deviations. In the text they are quoted with two significant digits; in the tables value(error) follows the PDG rounding (two digits if the leading digits are 10–35, else one). Bounds in the text ("at most", "agree to", "below") are rounded up, so they can exceed the corresponding table entry by one unit of the last digit. Monte Carlo values and the coefficients C_7, C_8 are quoted as their sources give them.

2 The problem and what is known

2.1 The long-range Ising chain and its thermal dimension

The ferromagnetic long-range Ising (LRI) chain has spins $S_i = \pm 1$ on the sites of a line ($d = 1$) and couplings $J_{ij} \propto 1/|i-j|^{1+s}$. It has a finite-temperature transition for $0 < s \le 1$; the transition is mean-field for $s < 1/2$, of Kosterlitz–Thouless type at $s = 1$ [6, 7], and for $1/2 < s < 1$ its critical point is neither mean-field nor short-range [4, 5]. There the spin field has the exactly known dimension

$$\Delta_\sigma = \frac{d-s}{2} = \frac{1-s}{2}, \qquad s = 1 - 2\Delta_\sigma, \qquad \varepsilon \equiv 2s - d = 2s - 1 = 1 - 4\Delta_\sigma, \tag{1}$$

because long-range interactions are not renormalized [5]. The quantity of interest is the dimension of the energy operator, the leading $\mathbb{Z}_2$-even scalar,

$$\Delta_\varepsilon = d - \frac{1}{\nu} = 1 - y_t, \tag{2}$$

where ν is the correlation-length exponent and $y_t = 1/\nu$ the renormalization-group (RG) eigenvalue of the temperature.

Two limits are known analytically:

- $s \to 1/2^+$ **($\Delta_\sigma \to 1/4$, $\varepsilon \to 0$).** The fixed point becomes Gaussian, $\Delta_\varepsilon \to 1/2$, and Δ_ε has an expansion $\Delta_\varepsilon = \sum_n C_n \varepsilon^n$ with $C_0 = 1/2$, $C_1 = -1/6$ and higher coefficients known through C_8 (Sec. 8.2, Ref. [1]). The series is asymptotic, but close to $s = 1/2$ it is far more precise than any numerical method. The leading irrelevant coupling (φ^4) has the correction exponent $\omega = \varepsilon + O(\varepsilon^2)$, so corrections to scaling decay ever more slowly as $s \to 1/2$.

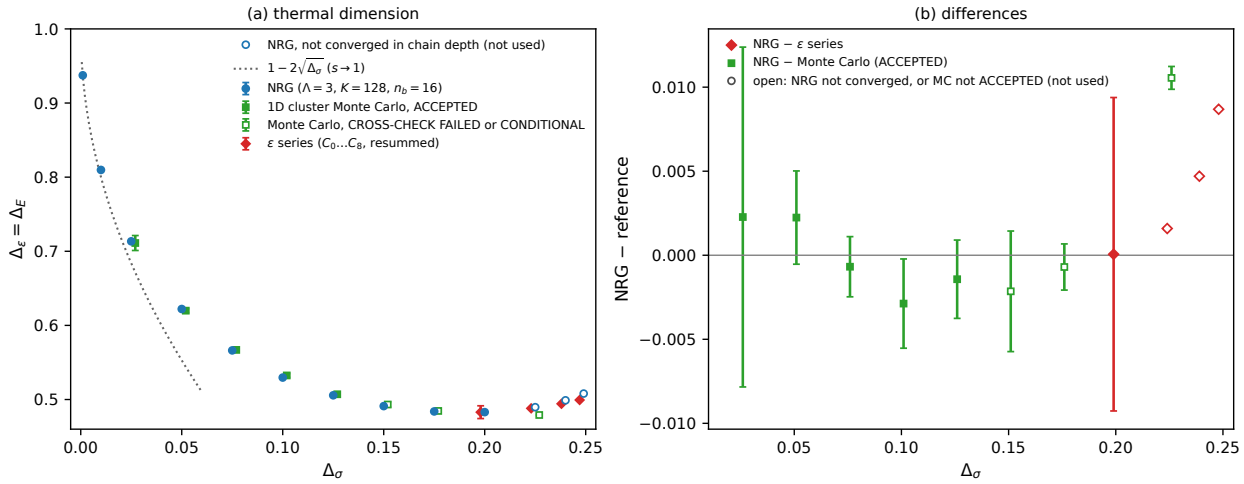


Figure 1: (a) Δ_E against Δ_σ: NRG (filled circles with $\delta_{\rm tot}$; open: not converged in chain depth, not used), the project's 1D cluster Monte Carlo study (filled squares: label ACCEPTED; open: CROSS-CHECK FAILED or CONDITIONAL, which that study does not certify as infinite-size values; bars $\sigma_{\rm stat}+\sigma_{\rm sys}$, added linearly), the resummed ε series (diamonds; bars: spread over Padé approximants, at $\Delta_\sigma=0.249$ the partial-sum error) and $1-2\sqrt{\Delta_\sigma}$ near $s=1$. (b) NRG minus reference; bars are $\delta_{\rm tot}$ plus the reference's error, added linearly. Open symbols are not used in any comparison.

- $s\to1^-$ **($\delta\equiv1-s=2\Delta_\sigma\to0$).** The two lowest $\mathbb{Z}_2$-even dimensions are $\Delta_\pm=1\pm\sqrt{2\delta}+\delta/4+O(\delta^{3/2})$ [8, 9]; the leading term is Kosterlitz's [6]. With $\Delta_-\sim\varphi^2$ the energy operator [8],

$$y_t=\sqrt{2\delta}-\frac{\delta}{4}+O(\delta^{3/2}),\qquad \omega=\Delta_+-1=\sqrt{2\delta}+\frac{\delta}{4}+O(\delta^{3/2}). \tag{3}$$

  So $\omega\approx y_t$ near $s=1$: both are small, but they are comparable (Sec. 3.9). Beyond this order the dual description gives Δ_ε as a series in $\sqrt{\Delta_\sigma}$,

$$\Delta_\varepsilon=1+\sum_{n\ge1}C_n^{\rm SR}(1)\,\Delta_\sigma^{n/2},\qquad C_1^{\rm SR}(1)=-2,\quad C_2^{\rm SR}(1)=\tfrac12, \tag{4}$$

  equivalent to Eq. (3) through its second term, with $C_3^{\rm SR}(1)$ to $C_8^{\rm SR}(1)$ computed in the accompanying paper [1]; $C_3^{\rm SR}(1)=\frac{39}{16}+2\zeta(2)$, $C_4^{\rm SR}(1)=4-4\zeta(2)-8\zeta(3)$, $C_5^{\rm SR}(1)=\frac{11249}{1024}-\frac{361}{16}\zeta(2)+24\zeta(3)+\frac{49}{2}\zeta(4)$ and $C_6^{\rm SR}(1)=26-38\zeta(2)+104\zeta(3)-80\zeta(4)-272\zeta(5)+112\zeta(2)\zeta(3)$.

2.2 The spin–boson model

The spin–boson model without a static bias field describes a two-level system coupled to a bath of harmonic oscillators [10]:

$$H=-\frac{\Omega}{2}\,\sigma_x+\sum_j\omega_j\,b_j^\dagger b_j+\frac{\sigma_z}{2}\sum_j\lambda_j\left(b_j+b_j^\dagger\right),\qquad \rho(\omega)\equiv\sum_j\lambda_j^2\,\delta(\omega-\omega_j)=2\alpha\,\omega^s\quad(0<\omega\le1). \tag{5}$$

Here $\sigma_{x,z}$ are Pauli matrices, Ω is the tunneling amplitude (often called Δ in the literature), b_j annihilates a bath oscillator of frequency ω_j, λ_j is its coupling, $\rho(\omega)$ is the bath spectral density (the convention of Ref. [12], equal to $J(\omega)/\pi$ in Ref. [10]), α is the dimensionless dissipation strength and $0<s<1$ ("sub-ohmic"). The bath cutoff frequency is the unit of energy, and $\Omega=0.1$ throughout; universality in Ω was not tested (Sec. 9).

For $0<s<1$ the model has a quantum phase transition at a critical coupling $\alpha_c(s,\Omega)$ [12, 16]. For $\alpha<\alpha_c$ (delocalized phase) the ground state is unique; for $\alpha>\alpha_c$ (localized phase) it is twofold degenerate, the spin being trapped in one of the eigenstates of σ_z. In the σ_z basis the imaginary-time path integral is a classical Ising chain along the imaginary-time axis, with lattice spacing of order the inverse cutoff [10, 16]: the tunneling Ω gives a short-range coupling between neighboring time slices, and integrating out the bath gives a long-range coupling $\propto\alpha/\tau^{1+s}$, so α plays the role of the inverse temperature. The quantum critical point is therefore in the universality class of the $d=1$ LRI model with the same s: $\langle\sigma_z(\tau)\sigma_z(0)\rangle\propto\tau^{-2\Delta_\sigma}$ there, and $\alpha-\alpha_c$, which is $\mathbb{Z}_2$-even, couples to the energy operator, so its RG eigenvalue is the y_t of Eq. (2). The other even coupling, Ω, feeds the same single relevant even direction.

2.3 The thirteen points

The study covers the thirteen values of Δ_σ in Table 1, chosen for a point-by-point comparison with other methods of the project. The outer points $\Delta_\sigma = 0.249$ ($s = 0.502$) and 0.001 ($s = 0.998$) are the hardest for every method: near $s = 1/2$ the fixed point is almost Gaussian and the approach to it is slow, near $s = 1$ the exponent y_t is small.

3 Method: bosonic NRG with an exact α-derivative

NRG [11, 14], extended to bosonic baths in Refs. [12, 13], solves an impurity coupled to a bath by mapping the bath onto a chain whose sites carry exponentially decreasing energy scales, and then diagonalizing the chain one site at a time, keeping only the lowest states. We use the standard formulation; the non-standard elements are the way y_t is extracted (Secs. 3.5–3.7).

3.1 Logarithmic discretization

The bath interval $(0, 1]$ is divided into bins $I_m = (\Lambda^{-(m+1)}, \Lambda^{-m}]$, $m = 0, 1, 2, \ldots$, with a discretization ratio $\Lambda > 1$. Each bin is replaced by a single oscillator c_m that carries the bin's spectral weight at its mean frequency [12, 13]. With $q = 1/\Lambda$, this "star" representation has

$$\gamma_m^2 = \int_{I_m} \rho(\omega)\, d\omega = \frac{2\alpha\,(1-q^{s+1})}{s+1}\, q^{m(s+1)}, \qquad \xi_m = \frac{\int_{I_m} \omega\rho(\omega)\, d\omega}{\gamma_m^2} = \xi_0\, q^m, \qquad \xi_0 = \frac{s+1}{s+2}\,\frac{1-q^{s+2}}{1-q^{s+1}}, \quad (6)$$

where γ_m is the coupling and ξ_m the frequency of oscillator m. Every decade of frequency gets the same number of modes, however low; dropping all other modes of a bin is the approximation of this step, which becomes exact as $\Lambda \to 1$.

3.2 Wilson chain

An orthogonal change of basis among the c_m (Lanczos tridiagonalization, started from the combination $\sum_m \gamma_m c_m$ that couples to the spin) maps the star onto a semi-infinite chain:

$$H = -\frac{\Omega}{2}\sigma_x + \frac{\lambda_{\mathrm{tot}}}{2}\,\sigma_z\,(b_0 + b_0^\dagger) + \sum_{n\ge 0} e_n\, b_n^\dagger b_n + \sum_{n\ge 1} t_{n-1}\,(b_{n-1}^\dagger b_n + b_n^\dagger b_{n-1}), \qquad \lambda_{\mathrm{tot}}^2 = \sum_m \gamma_m^2 = \frac{2\alpha}{s+1}. \quad (7)$$

The spin now couples only to site 0, and each site only to its neighbors; e_n are the on-site energies and t_n the hoppings. The star has weights $\gamma_m^2 \propto (aq)^m$ at the nodes $\xi_0 q^m$, with $a = q^s$: this is the orthogonality measure of the little q-Jacobi polynomials $p_n(x; a, 1|q)$ [23], so the recurrence coefficients are known in closed form. With

$$\mathcal{A}_n = \frac{(1-aq^{n+1})^2}{(1-aq^{2n+1})(1-aq^{2n+2})}, \qquad \mathcal{B}_n = \frac{a\,(1-q^n)^2}{(1-aq^{2n})(1-aq^{2n+1})},$$
$$\bar{e}_n \equiv \Lambda^n e_n = \xi_0\,(\mathcal{A}_n + \mathcal{B}_n), \qquad \bar{t}_n \equiv \Lambda^n t_{n-1} = \xi_0\,\sqrt{\mathcal{A}_{n-1}\mathcal{B}_n/q} \quad (n \ge 1). \quad (8)$$

(cf. also the orthogonal-polynomial chain mappings of Ref. [22]). The code evaluates these closed forms (`lri_spin.config.chain_coefficients`); the independent implementation of Sec. 8.1 obtains the same numbers by a high-precision Lanczos recursion. Both rescaled coefficients tend to constants, so site n carries the energy scale Λ^{-n}. A plain Lanczos recursion on the star is unstable even in multiple-precision arithmetic: without reorthogonalization it loses the coefficients after 13 sites with 49 digits and after 33 sites with 268 digits (`independent_check/NOTES.md`). The closed form avoids the problem.

3.3 Iterative diagonalization and truncation

Let H_n be the chain (7) truncated to the sites $0, \ldots, n$ and multiplied by Λ^n. Adding one site is the step

$$H_n = \Lambda\, H_{n-1}\otimes \mathbb{1} + \bar{e}_n\, \mathbb{1}\otimes b^\dagger b + \bar{t}_n\,\big(\bar{b}_{n-1}^\dagger \otimes b + \bar{b}_{n-1}\otimes b^\dagger\big), \qquad H_0 = -\frac{\Omega}{2}\sigma_x + \bar{e}_0\, b_0^\dagger b_0 + \frac{\lambda_{\mathrm{tot}}}{2}\sigma_z\,(b_0 + b_0^\dagger), \quad (9)$$

where $\mathbb{1}$ is the identity, $\bar{b}_{n-1}$ is the annihilation operator of site $n-1$ written in the basis kept at step $n-1$, and b acts on the new site. The factor Λ brings the previous levels to the new energy scale, so the levels of H_n stay of order one and their evolution with n is an RG flow; its plateaus are fixed points.

Three truncation parameters make each step finite:

- **Oscillator cutoff n_b.** Each new oscillator keeps its n_b lowest number states. This suffices near the delocalized and critical fixed points, where the chain sites hold few bosons [13].
- **Parity sectors.** The model conserves the parity $P = \sigma_x\,(-1)^{\sum_n b_n^\dagger b_n}$, which implements the Ising symmetry. H_n is diagonalized separately in the even and odd sectors, each of dimension at most Kn_b.
- **Kept states K.** In each sector the K lowest eigenstates are kept and the rest discarded. After each step all energies are shifted so that the ground state is at zero.

Together with Λ and the number N of chain sites (the last site sits at energy scale Λ^{-N}), these parameters define a finite numerical map from α to a sequence of spectra, which we call the *finite map*. Dense diagonalization costs of order $(Kn_b)^3$ per sector and site, which limits K and n_b.

3.4 Parity-gap flow and critical coupling

The observable is the rescaled parity gap

$$g_n(\alpha) = E_{0,n}^{\mathrm{odd}} - E_{0,n}^{\mathrm{even}}, \tag{10}$$

the difference between the lowest levels of H_n in the odd and even sectors. In the delocalized phase g_n flows to its value in the decoupled chain, where the spin is frozen by Ω and g_n is the lowest one-boson level; in the localized phase the ground state becomes degenerate and $g_n \to 0$; at α_c it stays on the value of the critical fixed point over an intermediate range of n.

For a chain of N sites, $n = 0, \ldots, N-1$, the critical coupling of the finite map is located by bisection on the sign of

$$f(\alpha) = g_{N-1}(\alpha) - 0.3\, g_{N-1}(0), \tag{11}$$

the last gap measured against 30% of the last gap of the decoupled chain ($\alpha = 0$) with the same parameters; $f > 0$ on the delocalized side and $f < 0$ on the localized side. The rule is:

- **Bracket.** Start from $[0.01, 2]$; halve the lower end (up to 40 times) until $f > 0$; double the upper end until $f < 0$, never evaluating $\alpha > 100$. Without a sign change the item fails.
- **Bisection.** At most 48 steps, stopping when the bracket is narrower than $5 \times 10^{-14} \max(1, |\alpha|)$. A bisection that does not reach this width is rejected.
- **Result.** α_c is the midpoint of the final bracket. The *terminal residual* is $f(\alpha_c)$.

This follows the script of the original study, with two conservative differences: the original evaluated α up to 128 and accepted an exhausted bisection. The fraction 0.3 is a convention, like the level used to define the crossover scale in Refs. [12, 13]. The value of α_c is not universal: it depends on Λ, K and n_b (Fig. 6) and enters no result.

3.5 Exact α-derivative of the truncated map

A small detuning $\delta\alpha = \alpha - \alpha_c$ points along the relevant direction of the critical fixed point and grows by a factor Λ^{y_t} per site, so near the fixed point

$$\partial_\alpha g_n\big|_{\alpha_c} \propto \Lambda^{y_t n}. \tag{12}$$

The production code measures this growth without finite differences, by differentiating every step (9) with respect to α, including the change of the kept states. In each parity block write $H = U\,\mathrm{diag}(E)\,U^{\mathrm{T}}$, with kept eigenvectors U_k (energies E_k) and discarded eigenvectors U_d (energies E_d), and let $\dot H = \partial_\alpha H$. In the gauge without rotations among kept states,

$$\dot U_k = U_d\, Z, \qquad Z_{ij} = \frac{\left(U_d^{\mathrm{T}} \dot H\, U_k\right)_{ij}}{E_{k,j} - E_{d,i}}, \qquad T = U_k^{\mathrm{T}} \dot H\, U_k - \dot E_0\, \mathbb{1}, \tag{13}$$

where $\dot E_0$ is the derivative of the ground-state energy (the shift to zero is part of the step). T, the derivative of the kept Hamiltonian, and the derivative of the projected site operator, $\dot{\bar b} = \dot U_k^{\mathrm{T}} B\, U_k + U_k^{\mathrm{T}} B\, \dot U_k$ with B the site-n annihilation operator in the unrotated product basis, are carried to the next step, where $\dot H_{n+1} = \Lambda\, T \otimes \mathbb{1} + \bar t_{n+1}(\dot{\bar b}^\dagger \otimes b + \mathrm{h.c.})$. Only H_0 depends on α explicitly, through $\lambda_{\mathrm{tot}} \propto \alpha^{1/2}$. The derivative of the gap is $\partial_\alpha g_n = T_{00}^{\mathrm{odd}} - T_{00}^{\mathrm{even}}$. This is the exact derivative of the finite map wherever the kept and discarded levels are separated; if the gap between them falls below $5 \times 10^{-13} \max(1, |E|)$ the item fails and no record is written. A finite difference in α would instead turn nonlinear once $\delta\alpha\, \Lambda^{y_t n}$ is no longer small, and lose digits to rounding as $\delta\alpha \to 0$.

3.6 The window audit

Near the fixed point, Eq. (12) gives $\ln|\partial_\alpha g_n| = \kappa + y_t\, n \ln\Lambda$. The window audit, inherited from the analyzer of the original study, is:

1. **Windows.** Windows of $w = \max(12, \lfloor 0.13N \rfloor)$ consecutive sites, with first sites from $\max(8, \lfloor 0.13N \rfloor)$ up to (excluding) $\lfloor 0.76N \rfloor - w$. A window is skipped if any gap $g_n \le 0$ or any $\ln|\partial_\alpha g_n|$ is not finite.
2. **Fit.** In each window a least-squares line in n is fitted to $\ln|\partial_\alpha g_n|$; its slope divided by $\ln\Lambda$ is y_t, and $\Delta_E = 1 - y_t$. The *fit residual* is the largest absolute deviation from the line.
3. **Flatness.** The *relative variation* of the window is $(\max g_n - \min g_n)/|\overline{g_n}|$.
4. **Quality windows** have relative variation < 0.005 and fit residual < 0.003 (inherited conventions). Their number is n_q, and their Δ_E values span the *quality range*.

The audit uses no target value of any exponent.

3.7 The estimate: settled value, plateau and extrapolation

The window values of an item are not constant (Fig. 7). Along the chain the flow first approaches the fixed point, with irrelevant corrections decaying as $e^{-\omega l}$ in the *RG time* $l = n\ln\Lambda$; once the residual detuning has grown (Sec. 3.9) it leaves the fixed point again. The estimate is taken where the flow is closest to the fixed point, by a rule with three steps (`code/analysis/estimator.py`):

1. **Settled value.** The *settled window* is the window with the smallest fit residual plus relative variation; its Δ_E is the *settled value*. This is the window the original analyzer reports, stored in every record as `selected`.
2. **Plateau test.** Take the windows whose first site lies within $h = \lceil w/4 \rceil$ sites of that of the settled window. The largest $|\Delta_E - \text{settled value}|$ among them is the *plateau drift*. A point has a *plateau* if the plateau drift of its central item is at most $\tau_\mathrm{p} = 5 \times 10^{-5}$. Its value is then the settled value, and the plateau drift is its chain-depth error.
3. **Extrapolation**, where there is no plateau. The local exponent

$$\Delta_\mathrm{loc}(n) = 1 - \frac{\ln|\partial_\alpha g_{n+1}| - \ln|\partial_\alpha g_{n-1}|}{2\ln\Lambda} \tag{14}$$

   is fitted with $\Delta_\infty + A\, e^{-\omega(l-l_0)}$ (free Δ_∞, A, ω; unweighted least squares) on $l_\mathrm{hi} - S \le l \le l_\mathrm{hi}$ for $S = 30, 25, 20$, where l_hi is the RG time of the last site of the settled window and l_0 that of the first fitted site. The value is the median of the three Δ_∞, and the chain-depth error is the larger of $|\text{extrapolated} - \text{settled}|$ and the spread of the three fits. Details: a range with fewer than 8 sites is skipped; the start values are Δ_∞ = last point, A = first minus last, $\omega = 0.2$; the fit is Levenberg–Marquardt. The extrapolation counts as *determined* only if every fit converged with $\omega > 0.01$; otherwise it is reported as not determined (n.d.), and a point that needs it would stop the analysis.

Every item of a point is evaluated with that point's estimator, so the scans of Sec. 4 compare like with like. The rule was written on 13 October 2026 and its constants were not varied afterwards (Sec. 4, App. C). It replaces the midpoint of the quality range used before; App. E explains why and lists the old values.

At $\Delta_\sigma = 0.001$ the rule's own fit ranges lie entirely on the flat plateau, so its extrapolation is not determined (n.d.); the point has a plateau and is not extrapolated.

Classification of the points. The rule puts each point in one of two classes, *plateau* or *extrapolated*. The eight points with $\Delta_\sigma \le 0.15$ have a plateau: their plateau drift is at most 1.91×10^{-5} (at $\Delta_\sigma = 0.15$) and at most 9.5×10^{-6} for $\Delta_\sigma \le 0.125$. None of the five points with $\Delta_\sigma \ge 0.175$ has one: their smallest drift is 7.74×10^{-5} (at $\Delta_\sigma = 0.175$). Every threshold between these two numbers gives the same classification. The drift grows smoothly with Δ_σ, so the split also depends on h and on the window width w, which the rule fixes. At $\Delta_\sigma = 0.15$ the drift is 6.6×10^{-5} with $h = w/2$ and 9.3×10^{-5} with windows of 20 sites, above τ_p in both cases, so the point would be extrapolated; every point with $\Delta_\sigma \le 0.125$ keeps its plateau (drifts below 2.2×10^{-5} and 1.6×10^{-5}). The values hardly depend on the class: extrapolating at $\Delta_\sigma = 0.15$ would move the value by -1.3×10^{-5}, windows of 20 sites move its settled value by $+1.8 \times 10^{-5}$, and taking the settled value at 0.175 would move it by $+8.6 \times 10^{-5}$, all well inside δ_tot. At the two final points without a plateau the extrapolation is the value (Table 4).

A third label, *not converged in chain depth*, is not an outcome of the rule. The author assigned it on 13 October 2026 to the three points closest to $s = 1/2$, $\Delta_\sigma = 0.249$, 0.24 and 0.225, when the comparison with the ε series was already known (App. C). The reasons: there $\omega\, l_\mathrm{max} \approx 61\varepsilon$ is small (Sec. 3.9); at $\Delta_\sigma = 0.249$ the excess over the series decays at a rate of the one-loop size (Sec. 7.1); and the excess of the settled values

over the series falls smoothly with ε (Sec. 8.2). At $\Delta_\sigma = 0.225$ the rule alone would extrapolate: its three fits converge, with $\omega = 0.059$–0.092, and give 0.487529 with $\delta_{\rm depth} = 2.1 \times 10^{-3}$. The label there rests on the trend from the two other points and on the Borel–Padé sum of the series; optimal truncation of the series, 0.490(4), does not exclude the settled value. For the three labeled points the settled value is shown, the extrapolations are cross-checks only, and no error is quoted. That they "enter no comparison" means that no agreement or disagreement is claimed from their values; the label itself came from a comparison.

3.8 Cross-check fits (not adopted)

The rule's fits end at the settled window and contain no term for the departure from the fixed point, so on a plateau they return the plateau almost by construction. Two fits that include that departure test the rule from outside (`code/analysis/crosscheck.py`, written for the second review round, App. C). Neither is adopted, and no value in this note is taken from them. Both fit $\Delta_{\rm loc}(l)$ of the central item with an approach term plus a *detuning term* $A_{\rm d}\, e^{(1-\Delta_\infty)(l-l_{\rm end})}$, whose growth rate $1-\Delta_\infty = y_t$ is that of the residual detuning (Sec. 3.9). Each fit range runs from a start $l_{\rm a}$ to $l_{\rm end} = l_N - k/y_t$, where $l_N = N \ln \Lambda$ is the end of the chain and y_t is taken from the settled value, so the last stretch of RG time, where the detuning dominates, is left out. The fits are Levenberg–Marquardt; the result is the median over the accepted fits, with their range and number $n_{\rm fit}$.

- **Late fit.** Approach term $A\, e^{-\omega(l-l_{\rm a})}$; $l_{\rm a} \in \{20, 25, 30, 35\}$ ($\{40, 50, 60, 70\}$ at $\Delta_\sigma = 0.001$), $k \in \{8, 10, 12\}$, at least 10 sites per range; a fit is accepted if its largest residual is below 2×10^{-6} (below 2×10^{-5} if no fit reaches 2×10^{-6}). It is computed at every point and shown for the final points (Table 2).
- **Crossover fit.** Approach term $A/(e^{\omega(l-l_0)} - 1)$, the one-loop crossover form of Eq. (17) with a free l_0, or $A\, e^{-\omega(l-l_{\rm a})}$; $l_{\rm a} \in \{13, 18, 22\}$, $k \in \{12, 15\}$, at least 8 sites per range; a fit is accepted if its largest residual is below 2×10^{-5} and $0 < \omega < 2$. It is computed at $\Delta_\sigma \geq 0.2$, where the approach is slowest (Tables 4 and 10).

The late fit against the rule. It reproduces the plateau values to 3×10^{-6} at $0.001 \leq \Delta_\sigma \leq 0.125$, and differs from the final values by -1.7×10^{-5}, $+1.4 \times 10^{-5}$ and $+6.9 \times 10^{-5}$ at $\Delta_\sigma = 0.15$, 0.175 and 0.2. This is inside $\delta_{\rm depth}$ except at $\Delta_\sigma = 0.001$, where it is $+2.9 \times 10^{-6}$ against $\delta_{\rm depth} = 1.7 \times 10^{-6}$, far inside $\delta_{\rm tot}$. At $\Delta_\sigma = 0.01$ its fitted $\omega = 0.197$ agrees with $\sqrt{2\delta} + \delta/4 = 0.205$ from Eq. (3). At $\Delta_\sigma = 0.2$ it suggests that the rule's extrapolation, which ends where the detuning already bends $\Delta_{\rm loc}$, lies slightly low, by about a tenth of $\delta_{\rm tot}$. The crossover fit is discussed with the ε series in Sec. 8.2.

3.9 Chain depth

The bisection stops when its bracket is narrower than 5×10^{-14} (relative), so the critical coupling of the finite map is located to $\delta\alpha \approx 5 \times 10^{-14}$. This width is at the rounding level of the map: the same item run on 1 and on 2 threads gives values of α_c that differ by up to 2.6×10^{-14}, and the two implementations of Sec. 8.1 differ by up to 7.2×10^{-14} (relative), so a narrower bracket would not locate the critical coupling better in double precision. The residual detuning also grows as $\Lambda^{y_t n}$, and the flow therefore stays near the fixed point only up to

$$n \lesssim n_{\rm max} \simeq \frac{\ln(1/\delta\alpha)}{y_t \ln \Lambda} \approx \frac{30.6}{y_t \ln \Lambda}, \qquad l \lesssim l_{\rm max} = n_{\rm max} \ln \Lambda \approx \frac{30.6}{y_t}, \tag{15}$$

with $30.6 = \ln(2 \times 10^{13})$: the usable RG time is fixed whatever Λ. The chain lengths follow this estimate: at $\Lambda = 3$, $N = 65$ for $s \leq 0.8$, then 70, 80, 90 and 105 for $s = 0.85$, 0.9, 0.95, 0.98, and $N = 250$ at $s = 0.998$; for other Λ, $N(\Lambda) = N(3) \ln 3 / \ln \Lambda$. Where $N \approx n_{\rm max}$ the terminal residual $|f(\alpha_c)|$ is of order 0.1–0.4, which is expected and not a failure. At $\Delta_\sigma = 0.025$, where $N = 90$ and $n_{\rm max} \approx 97$, the terminal residuals are 4×10^{-4} to 0.02; at $\Delta_\sigma \leq 0.01$ the chains are clearly shorter than $n_{\rm max}$ (at $s = 0.98$, $N = 105$ against $n_{\rm max} \approx 146$) and they are at most 2×10^{-4} at $\Delta_\sigma = 0.01$ and 4×10^{-5} at 0.001 (App. D).

The flow reaches its fixed point within the usable depth only if $\omega\, l_{\rm max} \approx 30.6\, \omega/y_t$ is large. Near $s = 1$, Eq. (3) gives $\omega/y_t \to 1$, so $\omega l_{\rm max} \approx 30$ however small y_t becomes. Near $s = 1/2$, $\omega \approx \varepsilon$ and $y_t \approx 1/2$, so $\omega l_{\rm max} \approx 61\varepsilon$: about 0.25, 2.4 and 6 at $\Delta_\sigma = 0.249$, 0.24 and 0.225. This is why the plateau is lost as $s \to 1/2$ and not as $s \to 1$. Extended-precision arithmetic would lengthen $l_{\rm max}$ in proportion to the number of digits.

4 Error budget

For each final point the value is that of Sec. 3.7 for the central item. Its error has three components, each computed with the point's estimator:

- δ_{depth} (chain depth): the plateau drift of the central item; at an extrapolated point, the larger of |extrapolated − settled| and the spread of the three fits (Sec. 3.7).
- δ_K (K and n_b): the spread (maximum minus minimum) over all $\Lambda = 3$ items with $K \geq 64$ and $n_b \geq 10$; at $\Delta_\sigma \geq 0.2$, where $n_b = 10$ is not converged (Sec. 7.2), items with $n_b = 10$ are left out.
- δ_Λ (discretization): the larger of two numbers.

  (a) The spread over the items at different Λ, each at the largest K run there, with equal n_b (16 at $\Delta_\sigma \geq 0.2$, else 10): $\Lambda = 2.5$ and 3 at $K = 128$ at every point; for $\Delta_\sigma < 0.2$ also $\Lambda = 4$, run only at $K = 64$; and $\Lambda = 2$ at $K = 256$ at $\Delta_\sigma = 0.1$ and 0.05. The K convergence of $\Lambda = 4$ at $K = 64$ is assumed, not tested: larger Λ needs fewer kept states (Sec. 7.3), and at $\Lambda = 3$ the values are converged at $K = 48$ already (Sec. 7.2).

  (b) The shift of the $\Lambda = 2.5$ value under its one doubling of K, from 64 to 128, at $n_b = 10$, the only n_b run at $\Lambda = 2.5$, $K = 64$ (at $\Delta_\sigma \geq 0.2$, $n_b = 10$ is not converged).

  Part (b) allows for the untested K convergence at $\Lambda = 2.5$. It is the larger part at 8 of the ten final points, so at most points the quoted error is mainly this allowance, not a measured systematic of the central value.

The total is the linear sum

$$\delta_{\mathrm{tot}} = \delta_{\mathrm{depth}} + \delta_K + \delta_\Lambda, \tag{16}$$

listed in Table 2 and shown in Fig. 2. The points that are not converged in chain depth have no δ_{depth} and no δ_{tot}. At $\Delta_\sigma = 0.2$, δ_K mostly measures how the extrapolation itself scatters between items (the detuning fits of Sec. 3.7 bring those items within about 2×10^{-5} of each other), so it partly counts δ_{depth} twice; this is conservative.

What this error is, and is not. None of the components is a standard deviation: δ_{depth} is a resolution or an extrapolation length, δ_K and δ_Λ are ranges of convergence scans. They are therefore added linearly, not in quadrature. Five choices are judgements, not measurements:

- the plateau threshold τ_{p}, the neighborhood h and the window width w (the classification is the same for every τ_{p} between 1.91×10^{-5} and 7.74×10^{-5}; for h and w see Sec. 3.7);
- the fit form, the fit ranges S and the end point l_{hi} of the extrapolation;
- taking the full extrapolation length as δ_{depth} at an extrapolated point;
- the allowance (b) in δ_Λ for one K doubling at $\Lambda = 2.5$;
- no allowance below $\Lambda = 2$: over $2 \leq \Lambda \leq 4$ the values agree to 1.9×10^{-5} at the two points where $\Lambda = 2$ is K-converged ($\Delta_\sigma = 0.1$, 0.05), with a small monotone rise with Λ far inside the error, but the limit $\Lambda \to 1$ itself is not tested (Sec. 7.3).

The window rule of Sec. 3.6 (width, range, thresholds) is inherited and was not varied in production. Its quality thresholds do not set the value, because the settled window passes any reasonable threshold. The window width does enter the plateau test, and with it the class of $\Delta_\sigma = 0.15$, but not its value beyond 2×10^{-5} (Sec. 3.7).

When the choices were made. The rule and its constants were fixed on 13 October 2026, after the NRG data, the earlier comparisons with the ε series and the Monte Carlo study, and the reviews' exploratory comparisons had been seen; no protocol was registered in advance. App. C gives the timeline.

5 Software and validation

5.1 Code

The solver is the Python package `lri_spin` (`code/lri_spin_boson_hpc/src/lri_spin`; NumPy, SciPy and threadpoolctl; float64 throughout).

- **Backend.** The `ordered` backend builds one parity block at a time in the basis order of the reference solver (`code/lri_spin_boson_hpc/reference/`) and reproduces its flows bitwise and its derivative to rounding (tested). The reordered `dense` and `banded` backends are experimental and were not used.
- **Workflows.** The critical search writes checkpoints every 8 sites and persists the bisection state, so an interrupted item resumes where it stopped.
- **Records.** Each item ends in one result record (App. A) carrying a SHA-256 checksum and an *identity*: the configuration, the kind of run, bracket and tolerances, the SHA-256 of the solver source and the

Table 2: Error budget. Columns: estimator (plateau, extrapolated, or settled for the points not converged in chain depth); Δ_E to six decimals; the chain-depth error δ_{depth}, the K/n_b spread δ_K, the discretization error δ_Λ and their linear sum δ_{tot}, in units of 10^{-4}; the number n_q of quality windows of the central item; the late fit of Sec. 3.8 minus Δ_E (cross-check, not adopted; shown for the final points). n.c.: not converged in chain depth.

Δ_σ	estimator	Δ_E	$10^4\delta_{\text{depth}}$	$10^4\delta_K$	$10^4\delta_\Lambda$	$10^4\delta_{\text{tot}}$	n_q	10^4 (late fit $-\Delta_E$)	status
0.249	settled	0.508030	n.c.	1.70	1.21	–	2	––	not converged
0.24	settled	0.498922	n.c.	5.12	2.54	–	4	––	not converged
0.225	settled	0.489668	n.c.	2.98	6.05	–	9	––	not converged
0.2	extrapolated	0.482980	4.12	2.47	0.49	7.08	15	+0.69	final
0.175	extrapolated	0.483900	0.86	0.80	1.80	3.46	19	+0.14	final
0.15	plateau	0.491106	0.19	0.28	0.68	1.15	21	−0.17	final
0.125	plateau	0.505672	0.01	0.15	0.38	0.54	23	−0.00	final
0.1	plateau	0.529608	0.02	0.05	0.27	0.33	23	−0.01	final
0.075	plateau	0.566260	0.01	0.05	0.24	0.30	28	−0.00	final
0.05	plateau	0.622115	0.00	0.03	0.23	0.27	34	−0.00	final
0.025	plateau	0.713422	0.01	0.02	0.26	0.29	38	+0.00	final
0.01	plateau	0.809747	0.09	0.02	0.35	0.47	41	+0.01	final
0.001	plateau	0.937424	0.02	0.04	1.05	1.11	106	+0.03	final

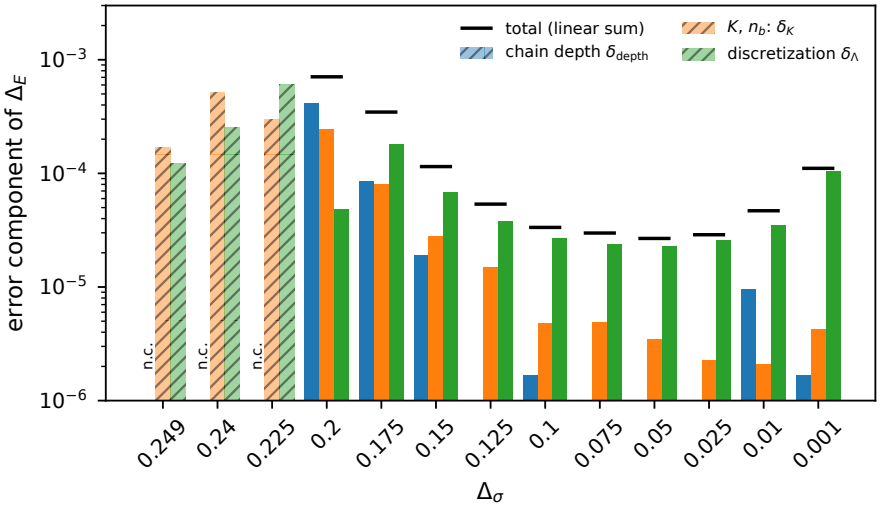


Figure 2: Components of the error at each Δ_σ and their linear sum δ_{tot}. Hatched: the three points not converged in chain depth, which have no δ_{depth} and no total. At the plateau points δ_Λ dominates, mostly through the K-doubling allowance at $\Lambda = 2.5$; at $\Delta_\sigma = 0.2$ the extrapolation length δ_{depth} dominates.

runtime (Python, NumPy, SciPy and OpenBLAS versions, BLAS kernel and threading layer). A record from another source or runtime is refused by the analysis.

Before production Claude Code read the package line by line against the reference solver (App. C); no numerical defect was found and no numerical code was changed.

5.2 Acceptance gates and tests

Every result record passed these gates before it was accepted: eigen-residual $\max|HU - UE| \le 10^{-8}$ and orthogonality error $\max|U^{\mathrm{T}}U - \mathbb{1}| \le 10^{-9}$ at sites $0, 1, 2$, every 16th site and the last; finite observables and sorted spectra; boson occupation within $[0, n_b - 1]$ and top-level probability (the weight of the highest kept level of the newest oscillator, p_{edge}) within $[0, 1]$; for derivatives, a positive gap between kept and discarded levels. Over all 197 production records the largest checked eigen-residual is 1.5×10^{-13} and the largest orthogonality error 5.0×10^{-13}.

The test suite (`make test`) has 85 tests. Physics: chain moments, small untruncated spectra against exact diagonalization, Kronecker products against block operators, reference trajectories, finite differences against the exact derivative. Software: workflows, record corruption, signals and locks, and the cluster tools of App. B (campaign ownership, generations, lanes, submission against a simulated Slurm, exports, stopping). Before production it passed on a compute node of the cluster for the final code, in the validation job that issued the receipt `inputs/validation_receipt/receipt.json`. The receipt binds the solver source to the node runtime: AMD EPYC 7543 nodes, Python 3.13.3, NumPy 2.1.2, SciPy 1.15.2, OpenBLAS 0.3.27/0.3.28 with

the Haswell kernel.

5.3 Reproducibility across machines

With the same runtime and thread counts, results are bitwise identical across nodes: a repeated smoke run on a second node reproduced the first bitwise (App. B). Across runtimes, rounding differences are amplified along the relevant direction as $\Lambda^{y_t n}$: at a *fixed* α, late flows of two machines differ, and even two implementations with bitwise identical α_c (configuration c1 of Sec. 8.1) have flows that differ by more than 10^{-8} from site 30 on. Archived values of α_c are therefore not portable, and every item retunes α_c on its own runtime. The critical coupling itself and the exponent are robust: the production values of α_c reproduce the three couplings archived by the original study on another machine to 7.1×10^{-14} (relative); the three example items, run on 2 threads, reproduce the α_c of the same items run on 1 thread bitwise in 2 cases and to 2.6×10^{-14} in the third; and the exponents of two implementations agree to 3.1×10^{-7} (Sec. 8.1).

6 Production campaigns

All items are critical searches followed by the exact derivative and the window audit, with $\Omega = 0.1$, bracket $[0.01, 2]$ and the `ordered` backend. Item IDs read `ds<`Δ_σ`>_L<`Λ`>_K<`K`>_B<`n_b`>_N<`N`>`; the example items use `s<`s`>_L<`Λ`>_K<`K`>_B<`n_b`>`. Items of the ds13 campaign and the Λ campaigns ran on 1 thread if $Kn_b \leq 1000$, 2 threads if $Kn_b < 1600$ and 4 otherwise; the example items ran on 2 threads, as in the package's example list.

Example (3 items). The package's example list: $s = 0.8$, 0.9, 0.98 at $K = 64$, $n_b = 10$, $\Lambda = 3$ (Table 3). They duplicate three items of the ds13 campaign, which makes them a test of thread-count independence (Sec. 5).

ds13 (146 items). All thirteen points at $\Lambda = 3$ with

- a scan in kept states, $K \in \{48, 64, 96, 128\}$ at $n_b = 10$;
- a scan in oscillator levels, $n_b \in \{8, 12, 16\}$ at $K = 64$;
- $\Lambda \in \{2, 4\}$ at $K = 64$, $n_b = 10$;
- the central item $K = 128$, $n_b = 16$;
- extras at the ends: $n_b \in \{20, 24\}$ for $\Delta_\sigma \geq 0.2$ and $K \in \{160, 192\}$ for $\Delta_\sigma \leq 0.05$.

Λ **pilot** (18 items). $\Delta_\sigma \in \{0.2, 0.1, 0.05\}$ at $\Lambda = 1.5$ ($K \in \{64, 128\}$ at $n_b = 10$, $K = 64$ at $n_b = 16$), $\Lambda = 2$ ($K = 128$) and $\Lambda = 2.5$ ($K \in \{64, 128\}$), $n_b = 10$ unless stated. ($\Lambda = 2$, $K = 64$ is part of the ds13 campaign.)

Λ **follow-up** (30 items). $\Lambda = 2.5$ at $K \in \{64, 128\}$ for the other ten points, plus $K = 128$, $n_b = 16$ for $\Delta_\sigma \geq 0.2$; and $\Lambda = 2$ at $K \in \{192, 256\}$ for $\Delta_\sigma \in \{0.2, 0.1, 0.05\}$.

The runs used 178 thread-hours (sum over items of wall time times threads) on whole cluster nodes. The allocations summed to 15 node-hours, i.e. about 970 core-hours as charged for 64-core nodes; the longest item took 2.7 h, and the production finished within two days including a cluster outage (Table 12 in App. B). The largest peak memory of an item was 495 MiB.

Table 3: The example items ($K = 64$, $n_b = 10$, $\Lambda = 3$, 2 threads): critical coupling, bisection steps, terminal residual $f(\alpha_c)$, number of quality windows, quality range and settled value. The terminal residuals near -0.16 reflect $N \approx n_{\max}$ (Sec. 3.9); at $s = 0.98$, $N < n_{\max}$.

item	α_c	bisection	terminal residual	quality windows	Δ_E range (settled)
`s0.80_L3_K64_B10`	0.43976978933704	46	-0.160	23	[0.52961, 0.53006] (0.52961)
`s0.90_L3_K64_B10`	0.68913573159160	46	-0.159	34	[0.62212, 0.62260] (0.62212)
`s0.98_L3_K64_B10`	1.08065976531605	46	$+0.000$	41	[0.80946, 0.81013] (0.80975)

7 Results

7.1 The approach to the fixed point

Figure 3 shows the local exponent (14) against the RG time, which is the quantity the estimator works on.

- **Plateau ($\Delta_\sigma = 0.1$, panel a).** After the transient, Δ_{loc} is flat over $30 \lesssim l \lesssim 55$ and the same at $\Lambda = 2$, 2.5, 3 and 4; the fitted correction exponent is $\omega = 0.339$. The settled window lies on the plateau.

- **No plateau ($\Delta_\sigma = 0.2$, panel b).** $\Delta_{\rm loc}$ is still falling when the detuning sets in. The settled value, 0.483392, is the deepest usable value; the extrapolation (Table 4; $\omega = 0.142$) gives 0.482980, -4.1×10^{-4} from it, and the extrapolations at $\Lambda = 2.5$ and 3 agree to 2×10^{-5}. At $\Delta_\sigma = 0.175$ ($\omega = 0.199$) the shift is -8.6×10^{-5}.
- **Near $s = 1/2$ ($\Delta_\sigma = 0.249$, panel c).** At one loop, with the φ^4 coupling u normalized so that $du/dl = \varepsilon u - u^2$ (fixed point $u^* = \varepsilon$), the local dimension of the energy operator is $\frac{1}{2} - \frac{\varepsilon}{2} + \frac{u}{3}$; the factor $\frac{1}{3} = (\mathcal{N}+2)/(\mathcal{N}+8)$ is that of the Ising case $\mathcal{N} = 1$ of the O($\mathcal{N}$) model [4], and at u^* it gives $\Delta_\varepsilon = \frac{1}{2} - \frac{\varepsilon}{6}$. A strong bare coupling, $u \to \infty$ at some RG time l_0, then gives

$$\Delta_{\rm loc} - \Delta_\varepsilon = \frac{\varepsilon/3}{e^{\varepsilon(l-l_0)} - 1}, \qquad \frac{1}{\Delta_{\rm loc} - \Delta_\varepsilon} = \frac{3}{\varepsilon}\big(e^{\varepsilon(l-l_0)} - 1\big), \tag{17}$$

  if $\Delta_{\rm loc}$ follows the running coupling. The inverse excess thus grows with local slope $3e^{\varepsilon(l-l_0)}$, which tends to 3 as $\varepsilon(l - l_0) \to 0$. With the series value $\Delta_\varepsilon = 0.499344$ the measured inverse excess grows nearly linearly in l at every K, n_b and Λ shown, with slope 2.55 for $15 \le l \le 25$, 2.87 over $20 \le l \le 45$ and 3.03 for $38 \le l \le 48$. Inferring l_0 from the data at $20 \le l \le 40$ gives $l_0 = 6$ to 9, for which Eq. (17) predicts local slopes 3.14–3.17 at $l = 20$ and 3.44–3.47 at $l = 43$. The measured slopes are 10–20% lower and rise with l as predicted. Terms of higher order in u, largest early in the flow where u is largest, could account for the difference, but we have not shown that. The test is semi-quantitative: it shows an approach of the one-loop kind, at a rate set by ε, not a precise coefficient. At $\Delta_\sigma = 0.249$, 0.24 and 0.225 the window Δ_E still changes by -2.5×10^{-4}, -2.0×10^{-4} and -1.9×10^{-4} per site at the settled window. Single-exponential fits give $\omega = 0.064$, 0.079 and 0.078, which approach ε (0.004, 0.04, 0.1) only at the largest ε, since an exponential cannot represent a $1/l$ approach, and the three fits scatter by up to 1×10^{-3} (Table 4). The data do not determine the fixed-point value at these points to the precision of the others: they are *not converged in chain depth*. A scan in K, n_b or Λ at fixed $l_{\rm max}$ cannot reveal this, because once K is converged and $n_b \ge 12$ the effect depends only on l ($n_b = 8$ and 10 are off by up to 9.5×10^{-3} and 8.9×10^{-4} near $s = 1/2$, Sec. 7.2).
- **Near $s = 1$ ($\Delta_\sigma = 0.001$, panel d).** $\Delta_{\rm loc}$ rises to a long plateau ($75 \lesssim l \lesssim 120$), as Eq. (15) allows. At the 10^{-4} level the plateau depends on the truncation away from $\Lambda = 3$: at $\Lambda = 2.5$ it moves by 1.05×10^{-4} between $K = 64$ and 128, at $\Lambda = 4$ ($K = 64$) it overshoots by about 10^{-4} before settling, and at $\Lambda = 2$, $K = 64$ is far from converged ($+5.6 \times 10^{-4}$). At the largest K run the values at $\Lambda = 2.5$, 3 and 4 agree to 2.3×10^{-5}, but at $\Lambda = 2.5$ that is a single doubling. The plateau value is therefore ambiguous at the 10^{-4} level; the budget carries this through the K-doubling allowance, which gives $\delta_\Lambda = 1.05 \times 10^{-4}$ at this point (Sec. 4).

Table 4: The points without a plateau (central items). Settled value; plateau drift; change of the window Δ_E per site at the settled window; the three fitted Δ_∞ and ω for $S = 30, 25, 20$; their median (the extrapolated value). At $\Delta_\sigma = 0.2$ and 0.175 the median is the final value. Last column: the crossover fit of Sec. 3.8 (median, number $n_{\rm fit}$ of accepted fits), computed at $\Delta_\sigma \ge 0.2$ only; a cross-check, not adopted. *Not converged in chain depth: cross-check only, not used.

				the rule (Sec. 3.7)			cross-check, not adopted
Δ_σ	settled	drift	per site	fits Δ_∞ ($S = 30, 25, 20$)	ω	median	crossover fit ($n_{\rm fit}$)
0.249*	0.508030	8×10^{-4}	-3×10^{-4}	0.50496, 0.50463, 0.50463	0.068, 0.064, 0.064	0.504628	0.50209 (6)
0.24*	0.498922	7×10^{-4}	-2×10^{-4}	0.49670, 0.49673, 0.49704	0.078, 0.079, 0.088	0.496734	0.49460 (6)
0.225*	0.489668	6×10^{-4}	-2×10^{-4}	0.48804, 0.48753, 0.48668	0.092, 0.078, 0.059	0.487529	0.48810 (7)
0.2	0.483392	2×10^{-4}	-7×10^{-5}	0.48305, 0.48298, 0.48291	0.151, 0.142, 0.130	0.482980	0.48305 (9)
0.175	0.483986	8×10^{-5}	-2×10^{-5}	0.48391, 0.48390, 0.48389	0.205, 0.199, 0.190	0.483900	–

7.2 Convergence in kept states and oscillator levels

Table 5 and Fig. 4 show the deviation of every $\Lambda = 3$ item from the central value.

- **Kept states.** The estimates are converged at $K = 48$ already: for $\Delta_\sigma \le 0.125$ the spread over K is at most 4.9×10^{-6}.
- **Oscillator levels.** n_b matters mainly near $s = 1/2$, where $n_b = 8$ is off by up to 9.5×10^{-3} and $n_b = 10$ by up to 8.9×10^{-4}, while at the final points all items with $n_b \ge 12$ (and $K \ge 64$) agree with the central value to 1.9×10^{-4}.
- **Top-level probability.** $p_{\rm edge}$ is the weight of the highest level, $n_b - 1$, of the newest oscillator in the ground state, evaluated at every site of the critical flow; App. D lists its maximum over the chain. That

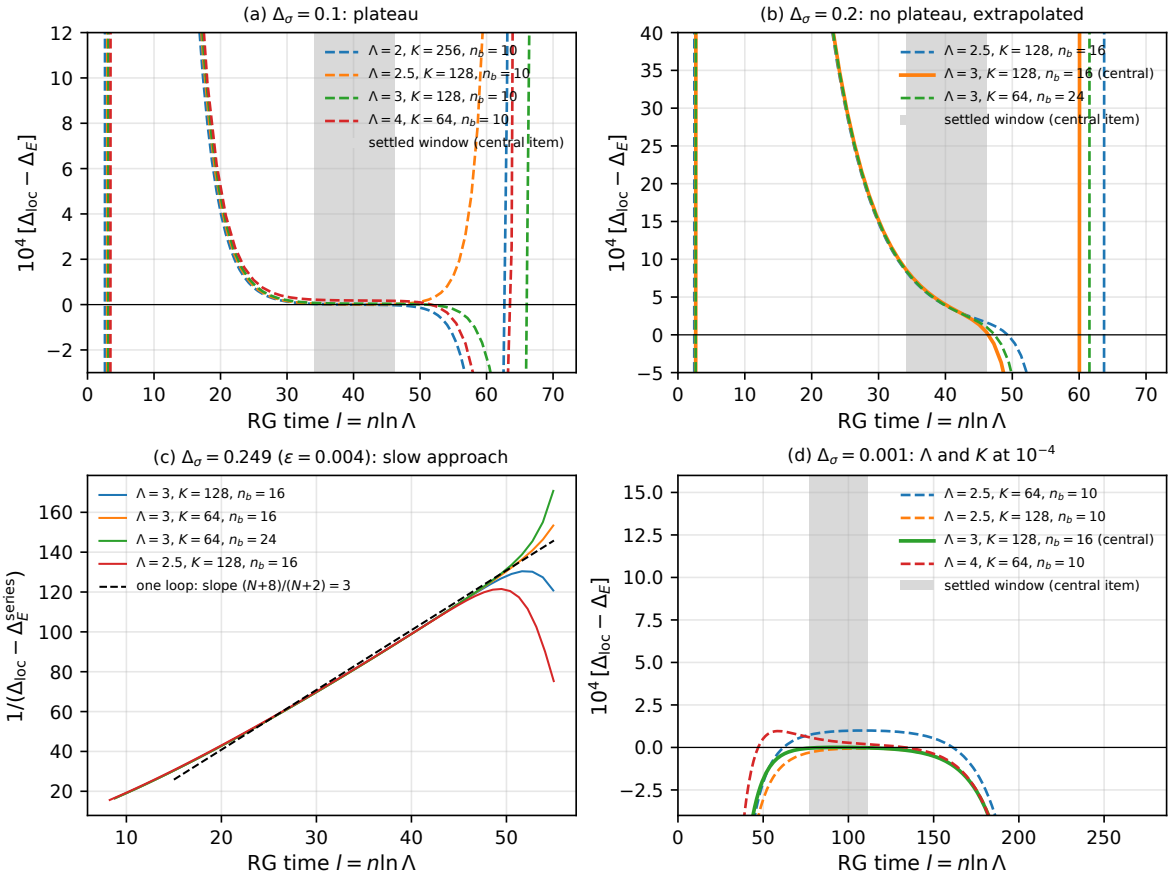


Figure 3: Local exponent $\Delta_{\rm loc}$, Eq. (14), against the RG time $l = n \ln \Lambda$. (a), (b), (d): $10^4\,(\Delta_{\rm loc} - \Delta_E)$ relative to the final value; gray band: the settled window of the central item. (a) $\Delta_\sigma = 0.1$, items at four values of Λ, each at the largest K run (at $\Lambda = 4$ only $K = 64$): a common plateau. (b) $\Delta_\sigma = 0.2$: no plateau; the zero line is the extrapolated value. (d) $\Delta_\sigma = 0.001$: Λ and K at the 10^{-4} level. (c) $\Delta_\sigma = 0.249$: inverse distance to the ε-series value against l, for four truncations; dashed: slope 3, the one-loop value for $\varepsilon(l - l_0) \to 0$ (Eq. (17)).

maximum is reached at the last sites, where the flow leaves the fixed point, and depends on the side it leaves to: for the items with $n_b \geq 12$ its median is 2×10^{-8} when the flow ends toward the localized phase (negative terminal residual; the bosons are displaced there) and 3×10^{-19} when it ends toward the delocalized phase. This is why it changes by many decades between items that differ only in K. For $n_b \geq 12$ it is at most 4.9×10^{-4}, at most 2.1×10^{-4} for the central items, and below 5.4×10^{-6} at $\Delta_\sigma \leq 0.15$. It is largest near $s = 1/2$, where the critical fixed point holds the most bosons.

7.3 Dependence on the discretization Λ

The chain length is scaled as $N \propto 1/\ln \Lambda$, so all items of a point reach the same RG time. The Λ studies separate truncation from discretization (Tables 6 and 7, Fig. 5):

- **Smaller Λ needs many more kept states.** At $\Lambda = 1.5$, going from $K = 64$ to 128 moves Δ_E by 0.028–0.040, and $K = 128$ is still 0.008–0.024 away from the converged $\Lambda = 2$ value; raising n_b to 16 changes Δ_E by only 1.9×10^{-6}. At $\Lambda = 2$, $K = 192$ and 256 agree to 4.7×10^{-5}. At $\Lambda = 2.5$, $K = 64$ and 128 (one doubling) agree to 6.8×10^{-5} for $0.01 \leq \Delta_\sigma \leq 0.15$, to 1.05×10^{-4} at 0.001, and differ by up to 1.6×10^{-4} at the final points with $\Delta_\sigma \geq 0.175$ (up to 6.1×10^{-4} at the points not converged in chain depth). At $\Lambda = 2$, $K = 64$ is not converged: it lies $+2 \times 10^{-4}$ to $+6 \times 10^{-4}$ above $\Lambda = 3$ at $\Delta_\sigma \leq 0.05$.
- **A small Λ dependence, far inside the error.** At the largest K run at each Λ (at $\Lambda = 4$ only $K = 64$, whose convergence is assumed), the plateau values at $\Lambda = 2$, 2.5, 3 and 4 agree to 1.9×10^{-5} at $\Delta_\sigma = 0.1$ and 0.05, and rise monotonically with Λ: relative to $\Lambda = 3$, $\Lambda = 2$, 2.5 and 4 lie at -3×10^{-6}, -2×10^{-6} and $+2 \times 10^{-5}$ at $\Delta_\sigma = 0.1$, and at -1×10^{-6}, -1×10^{-6} and $+5 \times 10^{-6}$ at 0.05. Extended to $\Lambda \to 1$ this would put the $\Lambda = 3$ value a few 10^{-6} above the limit, far below $\delta_{\rm tot}$. Over $\Lambda = 2.5$–4 the largest spread at a plateau point is 6.8×10^{-5} (at $\Delta_\sigma = 0.15$, set by $\Lambda = 4$). How many kept states the

Table 5: Deviation $10^4\,[\Delta_E(K, n_b) - \Delta_E^{\text{central}}]$ at $\Lambda = 3$ (central: $K = 128$, $n_b = 16$), each item evaluated with its point's estimator. In the scan over K, the offsets at $\Delta_\sigma \geq 0.2$ come from $n_b = 10$, not from K. Empty cells were not run.

	K scan ($n_b = 10$)						n_b scan ($K = 64$)					
Δ_σ	K=48	K=64	K=96	K=128	K=160	K=192	n_b=8	n_b=10	n_b=12	n_b=16	n_b=20	n_b=24
0.249	+10.66	+8.88	+10.28	+10.92			+94.49	+8.88	+1.08	−0.30	+1.24	−0.45
0.24	+7.03	+6.80	+10.23	+7.21			+89.86	+6.80	+3.00	+5.12	+0.70	+3.32
0.225	+2.43	+2.12	+4.70	+4.34			+65.41	+2.12	+0.20	+0.28	+1.88	−1.10
0.2	+2.07	+2.68	+3.46	+3.15			+33.99	+2.68	+1.84	+1.48	−0.63	−0.15
0.175	+0.81	+0.42	+0.80	+0.65			+16.97	+0.42	+0.20	+0.22		
0.15	+0.21	+0.21	+0.15	+0.17			+9.45	+0.21	−0.07	−0.07		
0.125	+0.15	+0.15	+0.13	+0.10			+5.86	+0.15	+0.03	+0.03		
0.1	+0.08	+0.05	+0.05	+0.04			+3.90	+0.05	+0.01	+0.03		
0.075	+0.06	+0.05	+0.03	+0.03			+2.86	+0.05	+0.02	+0.02		
0.05	+0.05	+0.03	+0.02	+0.02	+0.01	+0.01	+2.33	+0.03	+0.01	+0.02		
0.025	+0.04	+0.02	+0.02	+0.01	+0.01	+0.01	+2.26	+0.02	+0.01	+0.01		
0.01	+0.03	+0.02	+0.02	+0.02	+0.01	+0.01	+2.83	+0.02	+0.01	+0.01		
0.001	+0.04	+0.04	+0.04	+0.04	+0.03	+0.01	+7.08	+0.04	+0.00	+0.00		

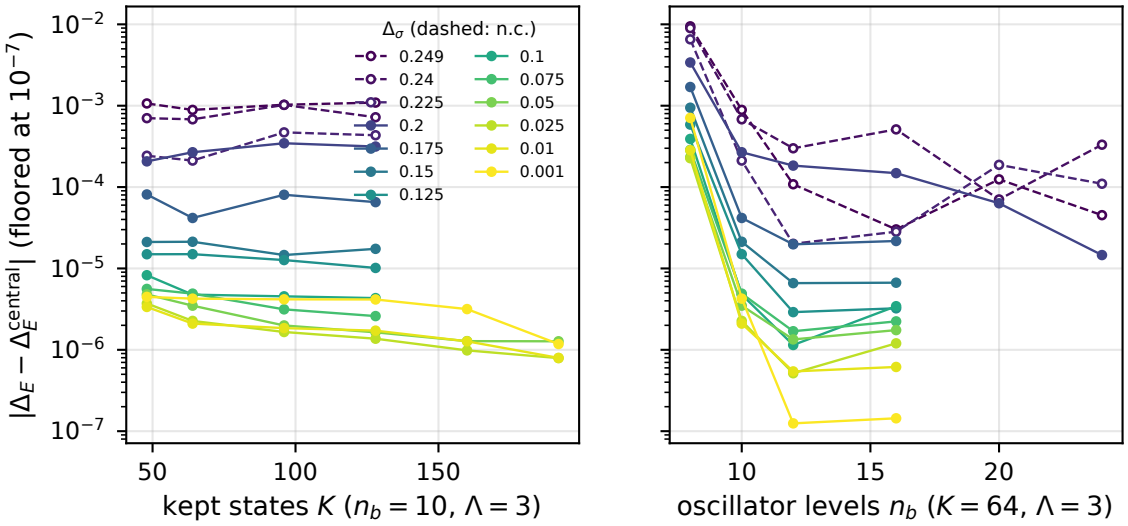


Figure 4: $|\Delta_E - \Delta_E^{\text{central}}|$ against K (left, $n_b = 10$) and n_b (right, $K = 64$) at $\Lambda = 3$; zero deviations are drawn at 10^{-7}. Flat curves mean converged. Dashed: the three points not converged in chain depth. In the left panel the level of the $\Delta_\sigma \geq 0.2$ curves is set by $n_b = 10$.

limit $\Lambda \to 1$ needs was not determined, and the limit is not tested.

- **At** $\Delta_\sigma = 0.2$ only $\Lambda = 2.5$ and 3 were run with the converged $n_b = 16$; their extrapolations agree to 2×10^{-5}. At $n_b = 10$ the extrapolated values spread by 4×10^{-4} over $\Lambda = 2$–4 (Fig. 5). We attribute this to an interplay of Λ with the $n_b = 10$ truncation (the $\Lambda = 2.5$ minus $\Lambda = 3$ difference changes sign between $n_b = 10$ and 16), but that is inferred, not measured; it is within δ_{tot} only through the large δ_{depth}.

An earlier version of this note reported that Δ_E rises as Λ decreases and covered the continuum limit with twice the $\Lambda = 2.5$ shift. Both came from the midpoint estimator: the window rule counts sites while the chain length scales as $1/\ln\Lambda$, so smaller Λ admitted a larger share of transient windows (App. E).

7.4 The final table

Table 1 gives the final values and Table 2 their errors. Starting from the Gaussian value 1/2 at $s = 1/2$, Δ_ε first decreases (the ε series at $\Delta_\sigma \geq 0.2$), passes through a shallow minimum of about 0.4830 near $\Delta_\sigma = 0.2$, and rises toward 1 as $s \to 1$. The window audits of three central items are shown in Fig. 7. The number of quality windows falls from 106 at $\Delta_\sigma = 0.001$ to 2 at $\Delta_\sigma = 0.249$ (Table 2).

7.5 The limit $s \to 1$

Table 8 compares $y_t = 1 - \Delta_E$ with the leading form $\sqrt{2\delta} = 2\sqrt{\Delta_\sigma}$ and the next-to-leading form of Eq. (3). The ratio to the leading form approaches 1 as $s \to 1$: $y_t = 0.06258$ against 0.06325 at $\Delta_\sigma = 0.001$ (1.1%) and

Table 6: The Λ study at three points ($n_b = 10$ unless stated): Δ_E with this note's estimator (plateau value at $\Delta_\sigma = 0.1$, 0.05; extrapolated at 0.2), the superseded midpoint with its half-width, and α_c. N is scaled with Λ so that the infrared scale Λ^{-N} is fixed. At $\Lambda = 1.5$ no item is converged in K.

Δ_σ	Λ	K	n_b	N	Δ_E (this note)	old midpoint (half-width)	α_c
0.2	1.5	64	10	176	0.536299	0.53659 (0.00036)	0.1626225790
0.2	1.5	64	16	176	0.536300	0.53660 (0.00036)	0.1626210299
0.2	1.5	128	10	176	0.507038	0.50754 (0.00102)	0.1603705735
0.2	2	64	10	103	0.484078	0.48534 (0.00177)	0.1676281822
0.2	2	128	10	103	0.483021	0.48471 (0.00165)	0.1676098637
0.2	2	192	10	103	0.483061	0.48448 (0.00193)	0.1676082707
0.2	2	256	10	103	0.483014	0.48449 (0.00188)	0.1676081346
0.2	2.5	64	10	78	0.483221	0.48426 (0.00131)	0.1761692489
0.2	2.5	128	10	78	0.483173	0.48440 (0.00101)	0.1761670918
0.2	3	64	10	65	0.483248	0.48404 (0.00063)	0.1847222031
0.2	3	128	10	65	0.483295	0.48406 (0.00061)	0.1847217711
0.2	4	64	10	52	0.483381	0.48401 (0.00034)	0.2008115424
0.1	1.5	64	10	176	0.567059	0.56714 (0.00019)	0.3786792319
0.1	1.5	64	16	176	0.567059	0.56714 (0.00019)	0.3786839607
0.1	1.5	128	10	176	0.539196	0.53968 (0.00049)	0.3733064368
0.1	2	64	10	103	0.529796	0.53042 (0.00063)	0.3936704020
0.1	2	128	10	103	0.529631	0.53025 (0.00063)	0.3936438035
0.1	2	192	10	103	0.529613	0.53024 (0.00063)	0.3936429089
0.1	2	256	10	103	0.529609	0.53023 (0.00064)	0.3936426886
0.1	2.5	64	10	78	0.529637	0.53010 (0.00046)	0.4167902745
0.1	2.5	128	10	78	0.529610	0.53007 (0.00046)	0.4167874486
0.1	3	64	10	65	0.529613	0.52984 (0.00022)	0.4397697893
0.1	3	128	10	65	0.529612	0.52984 (0.00023)	0.4397694449
0.1	4	64	10	52	0.529628	0.52981 (0.00018)	0.4825812310
0.05	1.5	64	10	217	0.670692	0.67093 (0.00026)	0.5909590331
0.05	1.5	64	16	217	0.670693	0.67092 (0.00026)	0.5909603112
0.05	1.5	128	10	217	0.630353	0.63061 (0.00026)	0.5780543351
0.05	2	64	10	127	0.622300	0.62262 (0.00032)	0.6120267877
0.05	2	128	10	127	0.622125	0.62241 (0.00036)	0.6119924970
0.05	2	192	10	127	0.622119	0.62244 (0.00033)	0.6119916485
0.05	2	256	10	127	0.622116	0.62242 (0.00034)	0.6119911289
0.05	2.5	64	10	96	0.622139	0.62240 (0.00026)	0.6507599139
0.05	2.5	128	10	96	0.622116	0.62235 (0.00029)	0.6507542509
0.05	3	64	10	80	0.622119	0.62236 (0.00024)	0.6891357316
0.05	3	128	10	80	0.622117	0.62236 (0.00024)	0.6891354544
0.05	3	192	10	80	0.622117	0.62236 (0.00024)	0.6891353149
0.05	4	64	10	63	0.622122	0.62223 (0.00013)	0.7603342269

0.19025 against 0.20000 at 0.01 (4.9%). Against the next-to-leading form 0.06275 the difference at $\Delta_\sigma = 0.001$ is -1.7×10^{-4}. The remainder divided by $\delta^{3/2}$ changes smoothly over $0.001 \le \Delta_\sigma \le 0.075$ (-1.90 to -1.32): it is -1.68 at $\Delta_\sigma = 0.01$, where $\delta_{\rm tot}/\delta^{3/2} = 0.02$, and -1.90 at 0.001, where $\delta_{\rm tot}/\delta^{3/2} = 1.2$ is as large as the entry. The dual expansion (4) fixes the coefficient of $\delta^{3/2}$ in y_t, $-C_3^{\rm SR}(1)/2^{3/2} = -2.025$, which the remainder approaches as $s \to 1$.

The accompanying paper [1] tests the dual expansion directly at $\Delta_\sigma = 0.01$, where $\delta_{\rm tot} = 4.7 \times 10^{-5}$. The difference between Δ_E and the series (4) summed through $C_n^{\rm SR}(1)$ is $+4.7 \times 10^{-3}$, -9.8×10^{-4}, $+2.4 \times 10^{-4}$ and -5.3×10^{-5} for $n = 2$, 3, 4 and 5: it shrinks to the size of $\delta_{\rm tot}$ as $C_3^{\rm SR}(1)$, $C_4^{\rm SR}(1)$ and $C_5^{\rm SR}(1)$ are added. The term of $C_6^{\rm SR}(1)$, of magnitude 5.9×10^{-5}, is itself of the size of $\delta_{\rm tot}$ and brings the difference to $+5.8 \times 10^{-6}$; the higher terms lie below $\delta_{\rm tot}$. NRG thus reproduces the dual expansion through $C_5^{\rm SR}(1)$ and is consistent with $C_6^{\rm SR}(1)$.

8 Checks

A second implementation of the same numerical map tests the code (Sec. 8.1); two methods that do not use NRG test the method (Secs. 8.2 and 8.3); Sec. 8.4 lists published values. Figure 1 summarizes the comparisons.

8.1 A second implementation

Design. Claude Code, which knew the production code, wrote a specification of the physics and conventions only (`independent_check/SPEC.md`). It fixed the discretization, the chain, the truncation per parity sector, the critical-search rule and the window rule, and forbade access to the production code, data and this note. From it a separate Claude agent wrote a second solver (`independent_check/indep_nrg.py`); that it did not open the forbidden files rests on the instruction and on the agent's own statement in `NOTES.md`, not on an access log. Every algorithmic choice differs from production:

Table 7: Λ and K shifts at all thirteen points, in units of 10^{-4}, relative to $\Lambda = 3$ at the same K and n_b ($\Lambda = 2.5$ uses $n_b = 16$ for $\Delta_\sigma \geq 0.2$, else 10). The $\Lambda = 2$, $K = 64$ column is not converged in K. Last columns: the K doubling at $\Lambda = 2.5$ ($n_b = 10$) and the number of quality windows of the $\Lambda = 2.5$, $K = 128$, $n_b = 10$ item.

	$10^4\times$ shift relative to $\Lambda = 3$			at $\Lambda = 2.5$	
Δ_σ	Λ=2, K=64	Λ=2.5, K=128	Λ=4, K=64	$10^4 \times$ (K=64 minus K=128)	n_q
0.249	+41.37	+1.21	+17.22	+0.88	13
0.24	+34.03	+0.71	+16.43	+2.54	13
0.225	+19.17	+2.29	+10.71	+6.05	17
0.2	+8.30	+0.18	+1.33	+0.49	22
0.175	+3.71	−0.53	+1.50	+1.55	25
0.15	+2.38	−0.10	+0.54	+0.67	28
0.125	+1.90	−0.07	+0.23	+0.38	31
0.1	+1.84	−0.02	+0.15	+0.27	32
0.075	+1.75	−0.02	+0.06	+0.24	36
0.05	+1.81	−0.01	+0.03	+0.23	43
0.025	+1.62	−0.01	+0.03	+0.26	47
0.01	+2.02	−0.00	+0.06	+0.35	49
0.001	+5.63	−0.11	+0.12	+1.05	126

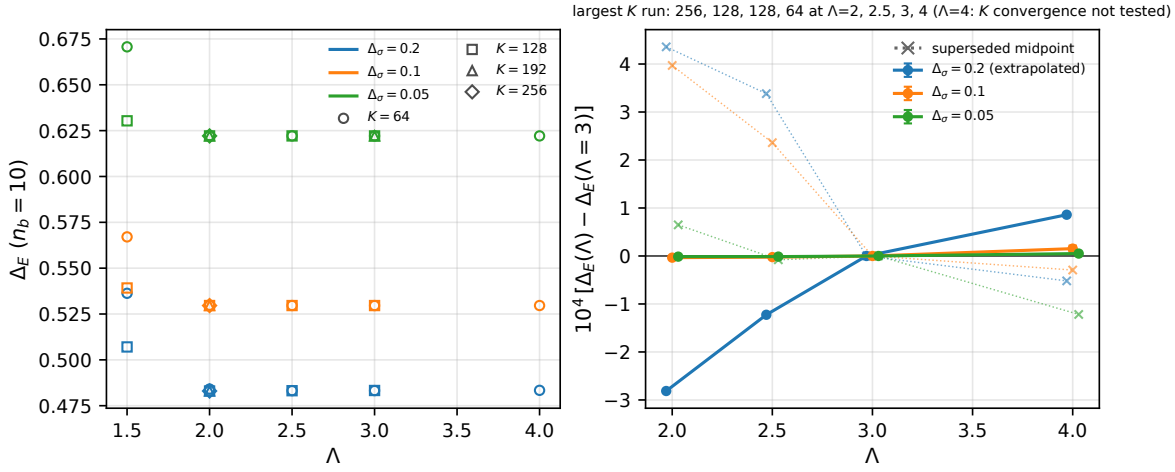


Figure 5: Left: Δ_E against Λ for all K at $n_b = 10$ (open symbols by K, color by Δ_σ); at $\Lambda = 1.5$ no item is converged in K. Right: K-converged items relative to $\Lambda = 3$ with this note's estimator (bars: plateau drift) and, dotted, the superseded midpoints, whose apparent Λ dependence is an artifact of the window rule.

	production (`lri_spin`)	second implementation (`indep_nrg.py`)
chain coefficients	closed form (8)	Lanczos with full reorthogonalization on the star, in mpmath
parity blocks	basis order of the reference solver	NumPy Kronecker order, generic operators
eigensolver	partial spectrum (flows), full (derivative)	full spectrum everywhere
y_t	exact derivative (13)	central finite differences in α, relative steps $\eta = 10^{-9}, 10^{-10}, 10^{-11}$
window audit	package code	written from the specification

The Lanczos recursion runs with $40 + 1.3N \log_{10}\Lambda$ decimal digits and is repeated with 40 more digits and 40 more star modes to check convergence. The second implementation has its own 38 self-tests (for example, the chain reproduces the star to 3×10^{-50}, and the NRG reproduces exact diagonalization of untruncated systems to 5×10^{-15}). Eight configurations c1–c8 with $0.502 \leq s \leq 0.998$, $\Lambda = 2$, 2.5 and 3, $K = 48$ or 64 and $n_b = 10$–16 ran on one cluster node, one single-core process each, the longest in 28 min; each is compared with the production record of the same item.

Result. Table 9 lists the comparison.

- The chain coefficients agree to better than 10^{-15} (relative): to rounding, whose last bits depend on the platform. The closed form is confirmed.
- The critical couplings are bitwise identical in 5 configurations and agree to 7.2×10^{-14} in the others. The critical flows agree to 5.1×10^{-13} over the first eleven sites.
- A window's finite-difference Δ_E is *trusted* when two neighboring steps η agree to 10^{-6}. In all 316 trusted windows (17 to 126 per configuration) the two implementations agree to 3.1×10^{-7}, with median

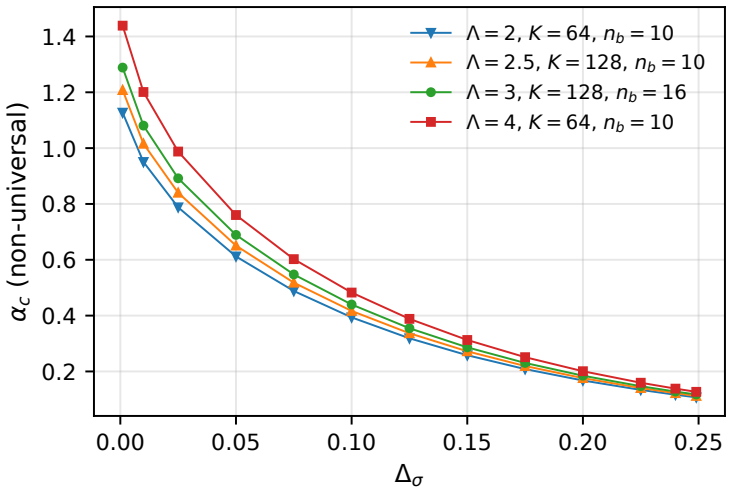


Figure 6: The critical coupling is not universal: it depends on Λ and weakly on K and n_b. It enters no result.

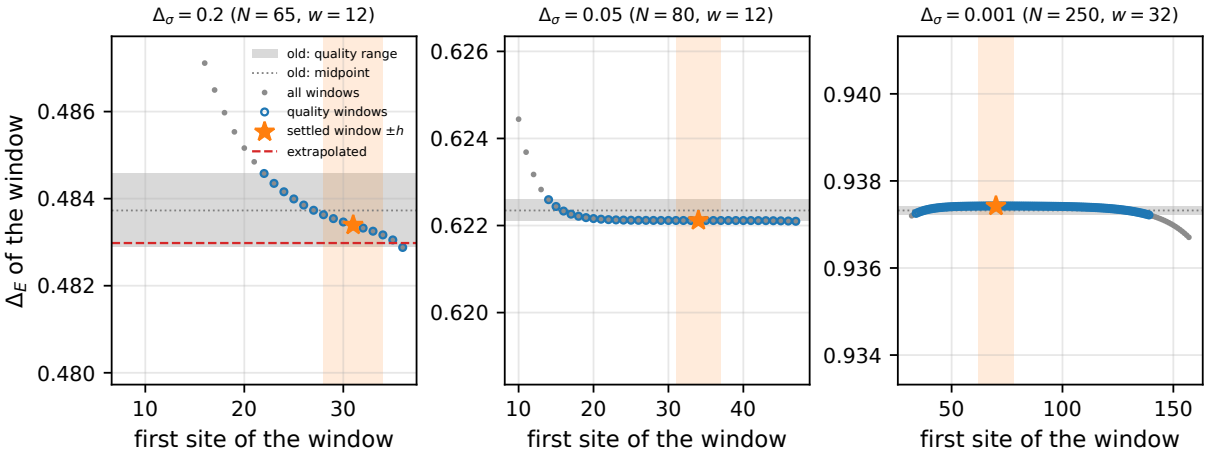


Figure 7: Window audit of three central items: Δ_E of every window (gray dots) and of the quality windows (open circles) against the first site of the window; star and orange band: the settled window and its neighborhood $\pm h$; gray band and dotted line: the quality range and its midpoint, the superseded estimate; dashed (at $\Delta_\sigma = 0.2$): the extrapolated value. The vertical scales differ between panels. At $\Delta_\sigma = 0.05$ and 0.001 the quality windows end on a flat plateau, at 0.2 they drift downward to the end.

4×10^{-9}, far below the quoted errors. In the 3 configurations whose settled window is trusted, they agree there to 8.2×10^{-8}.

- The latest windows of a chain lie beyond the reach of finite differences, because $\eta\, \alpha\, \Lambda^{y_t n}$ is no longer small there. Only 228 of the 296 production quality windows are within reach. All quality windows of a configuration are within reach only in $c3$ and $c6$; at $s = 0.502$ (c5) none is. This is a limit of the check, not a disagreement.

The check verifies the numerical map at $K \leq 64$. It does not test the central items at $K = 128$, $n_b = 16$ directly (they differ only in K and n_b), the estimator (it shares the window rule by construction), or the limit $\Lambda \to 1$. Both implementations come from the same model family, and the specification was written by the agent that knew the production code: the check tests the code, not shared conceptual choices.

8.2 The ε expansion near $s = 1/2$

Reference. The accompanying paper gives $\Delta_\varepsilon = \sum_{n=0}^{8} C_n \varepsilon^n$ for $d = 1$ [1], with $C_0 = 1/2$, $C_1 = -1/6$, $C_2 = 0.657131263981$, $C_3 = -3.719856322902$, $C_4 = 45.42488010356$, $C_5 = -731.13947662454$ and $C_6 = 14331.6870328$ (exact to the digits shown), $C_7 = -322550 \pm 40$ and $C_8 = (8.060 \pm 0.017) \times 10^6$. The uncertainties of C_7 and C_8 are the paper's allowances, mostly four sampling standard errors. They are not part of the Borel–Padé spread below; shifting C_7 or C_8 by about its allowance moves the [4/4] approximant by at most 1.1×10^{-5}

Table 8: $y_t = 1 - \Delta_E$ near $s = 1$ against the leading form $\sqrt{2\delta}$ ($\delta = 1 - s = 2\Delta_\sigma$) and the next-to-leading form $\sqrt{2\delta} - \delta/4$ of Refs. [8, 9]; the remainder in units of $\delta^{3/2}$, and $\delta_{\rm tot}$ in the same units (the uncertainty of the remainder).

Δ_σ	$\delta = 1 - s$	$y_t = 1 - \Delta_E$	$\sqrt{2\delta} = 2\sqrt{\Delta_\sigma}$	ratio	$\sqrt{2\delta} - \delta/4$	$(y_t - \sqrt{2\delta} + \delta/4)/\delta^{3/2}$	$\delta_{\rm tot}/\delta^{3/2}$
0.075	0.15	0.43374	0.54772	0.792	0.51022	-1.32	0.00051
0.05	0.1	0.37788	0.44721	0.845	0.42221	-1.40	0.00085
0.025	0.05	0.28658	0.31623	0.906	0.30373	-1.53	0.0026
0.01	0.02	0.19025	0.20000	0.951	0.19500	-1.68	0.017
0.001	0.002	0.06258	0.06325	0.989	0.06275	-1.90	1.2

Table 9: Second implementation against production. Columns: configuration (s, Λ, K, n_b, N); largest relative difference of the rescaled chain coefficients (a rounding-level bound); relative difference of α_c; number of trusted windows; largest window-by-window difference $|\delta\Delta_E|$ over them; production quality windows within finite-difference reach (covered/all); production quality range.

id	(s, Λ, K, n_b, N)	chain	α_c	trusted	max $\lvert\delta\Delta_E\rvert$	covered	quality range (production)
c1	$(0.8, 3, 64, 10, 65)$	$< 10^{-15}$	bitwise	19	2×10^{-7}	13/23	[0.529612, 0.530062]
c2	$(0.9, 3, 64, 10, 80)$	$< 10^{-15}$	4×10^{-14}	27	9×10^{-8}	23/34	[0.622119, 0.622597]
c3	$(0.98, 3, 64, 10, 105)$	$< 10^{-15}$	bitwise	53	3×10^{-7}	41/41	[0.809460, 0.810127]
c4	$(0.6, 3, 64, 12, 65)$	$< 10^{-15}$	bitwise	17	2×10^{-7}	3/15	[0.483316, 0.484567]
c5	$(0.502, 3, 64, 16, 65)$	$< 10^{-15}$	bitwise	18	2×10^{-7}	0/2	[0.508000, 0.508268]
c6	$(0.998, 3, 48, 10, 250)$	$< 10^{-15}$	4×10^{-14}	126	2×10^{-7}	106/106	[0.937223, 0.937429]
c7	$(0.8, 2, 64, 10, 103)$	$< 10^{-15}$	7×10^{-14}	34	2×10^{-7}	28/46	[0.529796, 0.531050]
c8	$(0.7, 2.5, 64, 10, 78)$	$< 10^{-15}$	bitwise	22	2×10^{-7}	14/29	[0.491178, 0.492328]

at $\Delta_\sigma = 0.225$ and 4.6×10^{-4} at 0.2, less than the spread at both points. The coefficients grow factorially, so the series is asymptotic. We evaluate it in two ways (`method_comparison/compare_methods.py`):

- **optimal truncation**: the partial sum up to, but excluding, the smallest term, with that term as its error;
- **Borel–Padé**: Padé approximants $[L/M]$ of the Borel transform $\sum_n C_n u^n/n!$ with $L + M = 6, 7, 8$, keeping those without a pole on the positive axis and with a value between 0.3 and 1.2, Laplace-transformed back. The central value is the upper of the two middle values of the $L+M = 8$ approximants (there are four at $0.2 \le \Delta_\sigma \le 0.24$), not their mean; the error is the half-range of all approximants with $L + M \ge 7$. The $L + M = 6$ approximants are computed but enter neither. The choice of the upper middle value is a convention of the comparison script of 8 October 2026, unchanged since and not chosen with the present NRG values in view; it has no further justification. At $\Delta_\sigma \ge 0.225$, where the approximants agree to within the quoted spread, it hardly matters; at $\Delta_\sigma = 0.2$ it does (below).

The reference is the partial sum at $\Delta_\sigma = 0.249$ ($\varepsilon = 0.004$), where all terms decrease and the error is 5×10^{-13} (Borel–Padé gives 0.499344 too), and the Borel–Padé sum at larger ε. At $\Delta_\sigma = 0.24$ the two evaluations agree to 3×10^{-5}. At 0.225 the Borel–Padé sum, 0.48808, has a small spread, but optimal truncation gives 0.490(4), so the reference there depends on the resummation. At $\Delta_\sigma = 0.2$ the spread, 9×10^{-3}, is too large to test NRG at the 10^{-3} level.

Result. Table 10. At $\Delta_\sigma = 0.2$ the series does not test NRG: the extrapolated NRG value lies $+6 \times 10^{-5}$ from the central approximant, a coincidence of the convention above, and $+2.1 \times 10^{-3}$ from the median 0.4809 of the 4 approximants with $L + M = 8$, which range from 0.468 to 0.483; both differences are well inside the spread, and neither is evidence for or against NRG. At the three points not converged in chain depth the settled values lie above the series by $+8.7 \times 10^{-3}$, $+4.7 \times 10^{-3}$ and $+1.6 \times 10^{-3}$, decreasing with ε, as the slow approach of Sec. 7.1 predicts. Two extrapolations serve as cross-checks; neither is adopted:

- **The rule's own extrapolation** (Sec. 3.7) is not stable there. Over the three fit ranges and $\Lambda = 2.5$, 3 it gives 0.4934–0.4970 (-8×10^{-4} to $+3 \times 10^{-3}$ from the series) at $\Delta_\sigma = 0.24$, and 0.4867–0.4888 (-1×10^{-3} to $+7 \times 10^{-4}$) at 0.225. At $\Delta_\sigma = 0.249$, where $\varepsilon l_{\max} \approx 0.25$, a single exponential cannot represent the $1/l$ approach, and the extrapolation (0.5046–0.5057) removes less than half of the excess.
- **The crossover fit** of Sec. 3.8 is stable and moves the values toward the series, but not onto it. At $\Delta_\sigma = 0.225$ it gives 0.48810 (7 accepted fits, 0.48808 to 0.48854, i.e. $+4 \times 10^{-6}$ to $+5 \times 10^{-4}$ from the series value 0.48808). At 0.24 it gives 0.49460 (0.49442 to 0.49501): every accepted fit lies above the series value 0.49422, by $+2 \times 10^{-4}$ to $+8 \times 10^{-4}$, against $+4.7 \times 10^{-3}$ for the settled value. At 0.249 it gives 0.50209, $+3 \times 10^{-3}$ from the series: there the fit is ill-conditioned, and the one-loop test of Sec. 7.1 is the relevant check.

Interpretation. The excess is the expected crossover from the strong bare coupling of the spin–boson model to the interacting fixed point, whose correction exponent $\omega \approx \varepsilon$ vanishes at $s = 1/2$; within $l_{\max}$ the flow has not arrived. It depends only on the RG time, not on K, n_b or Λ at fixed depth (Sec. 7.1), and at $\Delta_\sigma = 0.249$ it decays at a rate of the one-loop size. Logarithmic corrections of this kind were already seen in NRG at $s = 1/2$ [15]. An earlier version of this note attributed the excess to the known truncation errors of bosonic NRG near Gaussian fixed points (Hilbert-space truncation and mass flow [15, 17, 18]). The data do not support that: those errors affect order-parameter exponents below $s = 1/2$, and for $s > 1/2$ the mass-flow correction changes the flows only quantitatively (at $s = 0.6$, "no qualitative difference between the flows without and with mass-flow correction" [17]); we have no run that isolates either effect. The same mechanism acts at $\Delta_\sigma = 0.2$ and 0.175 at a smaller level (shifts -4.1×10^{-4} and -8.6×10^{-5} from settled to extrapolated), where the extrapolation is stable and its full length is in δ_{depth}.

Decision. The three values at $\Delta_\sigma = 0.249$, 0.24 and 0.225 are kept as computed (the settled values), marked "not converged in chain depth", shown in the tables and figures, and used in no comparison and no agreement statement. This is the author's decision. The label is not an outcome of the estimator, and it was assigned when this comparison was already known (Sec. 3.7); App. C gives its history, including an earlier recommendation that the author overruled.

Table 10: NRG against the ε series near $s = 1/2$. Optimal truncation with its smallest term; Borel–Padé central value with the spread over approximants; NRG minus the reference (the partial sum at $\Delta_\sigma = 0.249$, the Borel–Padé sum otherwise; at $\Delta_\sigma = 0.2$ this difference is not meaningful, see text); the range of the rule's extrapolations and the crossover fit of Sec. 3.8 (cross-checks, not adopted). Bracketed: settled values not converged in chain depth, not used.

Δ_σ	ε	NRG Δ_E	optimal truncation	Borel–Padé (spread)	NRG − reference	rule's extrapolation	crossover fit
0.249	0.004	[0.508030]	0.499344 ($\pm 5 \times 10^{-13}$)	0.49934 (6×10^{-15})	$+8.7 \times 10^{-3}$	0.5046–0.5057	0.50209
0.24	0.04	[0.498922]	0.494194 ($\pm 5 \times 10^{-5}$)	0.49422 (2×10^{-7})	$+4.7 \times 10^{-3}$	0.4934–0.4970	0.49460
0.225	0.1	[0.489668]	0.489905 ($\pm 4 \times 10^{-3}$)	0.48808 (1×10^{-4})	$+1.6 \times 10^{-3}$	0.4867–0.4888	0.48810
0.2	0.2	0.4830(7)	0.466667 (± 0.03)	0.48292 (9×10^{-3})	$+5.9 \times 10^{-5}$	adopted	0.48305

8.3 The 1D cluster Monte Carlo study

Reference. The Monte Carlo study of the same model in this repository (`MC/` folder [2]; Fukui–Todo cluster algorithm [24, 25] with finite-size scaling) is the project's own, run by the same AI systems; it is independent of NRG in method, not in authorship. We use only its quoted values and labels; their definitions are in its note [2]. In brief, its value of Δ_ε comes from a finite-size-scaling fit of all lattice sizes, with a statistical error σ_{stat} and a systematic scale σ_{sys}, the half-range of the values over a family of fit models (a sensitivity scale, not a standard deviation). As a cross-check it also extrapolates in the lattice size the crossings, between pairs of sizes, of a finite-size-scaling observable; this *crossing extrapolation* is quoted as $\Delta_\varepsilon^\infty(\sigma_\infty)(\Delta_{\omega'})$, with error σ_∞ and sensitivity $\Delta_{\omega'}$ to the assumed correction exponent ω' ($\Delta_{\omega'}$ is a sensitivity, not a dimension). Each value carries a label: **accepted**, the fit value and the crossing extrapolation agree within twice the error of their difference ("σ_{sys}-dominated" marks values whose σ_{sys} is much larger than σ_{stat}); **cross-check failed**, they differ by more, and the study does not certify the value as an infinite-size value; **cross-check unavailable**, fewer than two size series are available for the crossing extrapolation; **conditional**, σ_{sys} could not be estimated.

Result. Table 11.

- **Accepted ($\Delta_\sigma = 0.125, 0.1, 0.075, 0.05, 0.025$).** NRG and Monte Carlo differ by at most 2.9×10^{-3} (at $\Delta_\sigma = 0.1$). The differences exceed the Monte Carlo statistical errors, by up to $14\,\sigma_{\text{stat}}$, and have the same sign at neighboring points (below). They lie within σ_{sys} at 3 of the 5 points and within $\sigma_{\text{stat}} + \sigma_{\text{sys}} + \delta_{\text{tot}}$ at 4, the largest ratio being 1.08 at $\Delta_\sigma = 0.1$. This linear sum adds a standard deviation, a fit-model half-range and an NRG error scale; we use it as a scale for the size of the differences, not as a test of statistical consistency. At $\Delta_\sigma = 0.1$ the Monte Carlo crossing extrapolation agrees with NRG to $0.10\,\sigma_\infty$; at all 5 accepted points it agrees within $1.05\,\sigma_\infty$, and within 0.19 of $\sigma_\infty + \Delta_{\omega'} + \delta_{\text{tot}}$. The NRG errors are smaller than σ_{sys} by factors 36–333, so the comparison tests the Monte Carlo systematics more than it tests NRG; only σ_{sys} reconciles the two.
- **Cross-check failed ($\Delta_\sigma = 0.175$ and 0.15).** These are not infinite-size values, and we use them only to say where NRG lies: at 0.175 between the fit and the crossing extrapolation (-7.0×10^{-4} from the fit, $+8.6 \times 10^{-4}$ from the crossing), at 0.15 below both (-2.1×10^{-3} and -7.2×10^{-4}). Including these two points, the crossing extrapolations agree with NRG within $1.06\,\sigma_\infty$ at all 7 common points.

- **Correlated signs.** NRG lies below the Monte Carlo fit at 5 consecutive points of the 7 (0.175 to 0.075) and above it at 0.05 and 0.025. The differences are therefore not independent fluctuations; they look like a smooth systematic of one of the two methods, which the Monte Carlo $\sigma_{\rm sys}$ is meant to cover.
- $\Delta_\sigma = 0.225$ is conditional in the Monte Carlo study and not converged in NRG; the Monte Carlo value lies -9.0×10^{-3} from the series. It enters no comparison.

Table 11: NRG against the 1D cluster Monte Carlo study at the common points. MC value with $\sigma_{\rm stat}$ and $\sigma_{\rm sys}$ in units of the last digit, and its label (Sec. 8.3); NRG minus MC; "ratio" is |NRG − MC| divided by $\sigma_{\rm stat} + \sigma_{\rm sys} + \delta_{\rm tot}$ (shown also for the cross-check-failed points, which are not infinite-size values); MC crossing extrapolation with σ_∞ and $\Delta_{\omega'}$. MC values are quoted as that study gives them. *Not converged in NRG, not used.

Δ_σ	NRG Δ_E	MC Δ_ε $(\sigma_{\rm stat})(\sigma_{\rm sys})$	MC label	NRG − MC	ratio	MC crossing $(\sigma_\infty)(\Delta_{\omega'})$
0.225*	[0.4897]	0.47912(68)(–)	conditional (no $\sigma_{\rm sys}$)	$+0.011$	–	–
0.175	0.48390(35)	0.48460(35)(67)	cross-check failed	-7.0×10^{-4}	0.51	0.48304(81)(1)
0.15	0.49111(11)	0.49325(41)(306)	cross-check failed	-2.1×10^{-3}	0.60	0.49183(71)(70)
0.125	0.50567(5)	0.50710(35)(193)	accepted	-1.4×10^{-3}	0.61	0.50640(172)(392)
0.1	0.529608(33)	0.53248(20)(242)	accepted ($\sigma_{\rm sys}$-dom.)	-2.9×10^{-3}	1.08	0.52975(135)(248)
0.075	0.566260(30)	0.56694(37)(139)	accepted	-6.8×10^{-4}	0.38	0.56843(207)(1044)
0.05	0.622115(27)	0.61987(55)(220)	accepted	$+2.2 \times 10^{-3}$	0.81	0.62393(348)(773)
0.025	0.713422(29)	0.71115(49)(959)	accepted ($\sigma_{\rm sys}$-dom.)	$+2.3 \times 10^{-3}$	0.22	0.70906(902)(1509)

8.4 Published values

- **1D LRI Monte Carlo.** Uzelac, Glumac and Aničić [26] give $1/\nu = 0.50(2)$, 0.51(2), 0.47(2) and 0.40(1) at $s = 0.6$, 0.7, 0.8 and 0.9, i.e. $\Delta_\varepsilon = 0.50(2)$, 0.49(2), 0.53(2) and 0.60(1) at $\Delta_\sigma = 0.2$, 0.15, 0.1 and 0.05. NRG differs from them by -0.85, $+0.055$, -0.020 and $+2.2$ of their errors; at $\Delta_\sigma = 0.05$ the project's Monte Carlo value, 0.61987(55)(220), also lies above theirs. Angelini, Parisi and Ricci-Tersenghi [27] give $1/\nu = 0.4124(13)$ at $s = 0.875$ ($\Delta_\varepsilon = 0.5876(13)$ at $\Delta_\sigma = 0.0625$) and 0.506(14) at $s = 0.65453$; neither point is on our grid. Four interpolations of our final values (a cubic spline over $0.01 \le \Delta_\sigma \le 0.15$ and polynomials through three or four neighboring points) give 0.5898–0.5918 at $\Delta_\sigma = 0.0625$, 1.7 to 3.2 of their errors above 0.5876; the project's Monte Carlo study gives 0.59003(35)(150) there (label: cross-check unavailable).
- **The spin–boson model.** Bosonic NRG [12, 15] and continuous-time quantum Monte Carlo [20] give $1/\nu$ (or νz) for $s > 1/2$ only in figures. Read off the figures by eye, the NRG values of Ref. [15] are about 0.50, 0.50, 0.46, 0.37 and 0.28 at $s = 0.6$, 0.7, 0.8, 0.9 and 0.95, against our $y_t = 0.517, 0.509, 0.470, 0.378$ and 0.287, and the quantum Monte Carlo values agree with both within their error bars; this is agreement at the resolution of the figures, a few hundredths, not a quantitative test. Other methods for this model [21, 19] report order-parameter exponents, not ν.
- **Functional RG and the hierarchical model** [28] give $1/\nu$ for $1/2 < s < 1$ in figures; near $s = 1$ they quote the form of Eq. (3).

9 Limitations and open problems

1. **Chain depth near $s = 1/2$.** At $\Delta_\sigma = 0.249$, 0.24 and 0.225 the flow does not reach its fixed point within the RG time that double precision allows (Sec. 7.1). The crossover fit of Sec. 3.8 (cross-check, not adopted) is stable there and gives 0.48810 and 0.49460 at $\Delta_\sigma = 0.225$ and 0.24, closer to the series than the settled values but above it at 0.24; it is ill-conditioned at 0.249 (Sec. 8.2). Extended precision (longer $l_{\rm max}$) would remove the need to extrapolate. At $\Delta_\sigma = 0.2$ and 0.175 the values are extrapolations; they depend on the fit form, and $\delta_{\rm depth}$ covers the full extrapolation length.
2. **Continuum limit.** Over $2 \le \Lambda \le 4$ the plateau values change by at most a few 10^{-5}, monotonically, but $\Lambda < 2$ is not reached, K convergence at $\Lambda = 2.5$ rests on one doubling (covered by an allowance) and at $\Lambda = 4$ is assumed. At $\Delta_\sigma = 0.001$ the plateau depends on K at $\Lambda \ne 3$ at the 10^{-4} level. Runs at $\Lambda \le 1.5$ with more than 128 kept states, or at $\Lambda = 2$ with $K \ge 192$ at all points, would improve this. At $K = 512$ and $n_b = 10$ a parity block has dimension about 5000, which dense diagonalization handles, but it must be repeated at every site of every bisection step.
3. **Universality in Ω.** All runs use $\Omega = 0.1$. A second Ω at one or two points would test universality and the reading of the approach as a correction to scaling (the l dependence should move, the plateau value not).

4. **Error model.** The errors are scales, not standard deviations; the linear sum, the plateau threshold, the fit form and ranges, the full extrapolation length and the K-doubling allowance are judgements (Sec. 4). The estimator was chosen after the NRG data and after an earlier comparison had been seen, and the label "not converged in chain depth" was assigned after the comparison with the ε series.
5. **Coverage of the second implementation.** It ran at $K \leq 64$, could not reach the latest windows, including both quality windows at $s = 0.502$, and shares the window rule (Sec. 8.1).
6. **Review.** The analysis, the estimator and this note were reviewed only by AI agents: two rounds on 13 October 2026 by agents of the same family as the AI that wrote them, and a reading on 15 October (App. C). The changes made after the second round and after the reading have not been reviewed again.
7. **Not needed.** Hamiltonian-truncation calculations of the same model [1, 3] need a correction for a fixed energy cutoff on a free Fock space. NRG truncates iteratively in the interacting basis, keeps the complete new site for every kept state and is non-perturbative in α, so no such correction applies, and none was added.

10 Reproduction

10.1 The released folder

The `NRG/` folder of the repository contains:

- `code/lri_spin_boson_hpc/`: the solver, its tests, the reference solver of the original study and the cluster tools (App. C lists what was removed from the original tree);
- `code/campaigns/`: the generators of the item lists in `inputs/items/`;
- `code/analysis/`: the analysis stages (`summarize.py`, `estimator.py`, `build_tables.py`, `make_figures.py`, `note_numbers.py`, `check_figures.py`) and the collector of cluster records, `collect_records.py`;
- `inputs/`: the item lists, the frozen manifests that were submitted, the validation receipt and the environment record;
- `data/`: the 197 result records, the per-job lane logs and the job accounting;
- `results/`: the final table (`ds13_final.json`, `ds13_final_table.md`) and the Λ study;
- `independent_check/`, `independent_check_compare/`, `method_comparison/`: the checks of Sec. 8;
- `verify/`: the verifier, the expected values and the smoke-test reference; `slurm/`: job templates; `note/`: this note.

`README.md`, `REPRODUCE.md`, `PROVENANCE.md`, `REVIEWS.md` and `data/DATA.md` in that folder give the details.

10.2 How to run

From `NRG/`, with Python ≥ 3.10, NumPy, SciPy, threadpoolctl, mpmath, matplotlib and pytest:

- `make test` (about 75 s on one core of a recent laptop): the 85 solver tests, the checksum list of the code, and the 38 self-tests of the second implementation.
- `make replay` (20–30 s): checks the solver source hash, verifies every record (checksum, identity against its manifest and the receipt, numerical gates), recomputes all results, tables, in-text numbers and figures of this note in a hidden workspace `.work/`, and compares them with the released files and with `verify/expected.json`. It also needs the Monte Carlo table of the repository, `../MC/data/1d/scan_data/FIT_V3_SCAN.json` (`make replay MCSCAN=<path>` to point elsewhere). The extrapolations are least-squares fits whose last bits depend on the NumPy/SciPy build and the CPU, so the comparison uses tolerances (about 10^{-9} for values, one unit of the last printed digit for tables), not byte identity (`REPRODUCE.md`).
- `make smoke` (about 15 s): one small critical search with both solvers, compared with each other and with the reference values in `verify/smoke_expected.json`.
- `make note`: rebuilds this note in place from the released tables and figures.

Rerunning production needs a Slurm cluster: 178 thread-hours, about 15 node-hours of whole-node allocations (`REPRODUCE.md`). On another runtime the exponents are expected to agree within the quoted errors, while α_c moves at the 10^{-13} level and late flows at fixed α differ (Sec. 5); no full rerun on a second runtime was made.

References

[1] The accompanying paper on the long-range Ising model (in preparation): ε-expansion coefficients $C_0, \ldots, C_8$ for $d = 1$, with their derivations in the `GFF/` folder of this repository.

[2] *Monte Carlo determination of the thermal dimension $\Delta_\varepsilon(\Delta_\sigma)$ of the long-range Ising model in $d = 1, 2, 3$*, note in the `MC/` folder of this repository (2026).

[3] Notes on Hamiltonian truncation of the long-range Ising model, in the `HT/` folder of this repository (2026).

[4] M. E. Fisher, S.-k. Ma, and B. G. Nickel, *Critical exponents for long-range interactions*, Phys. Rev. Lett. **29**, 917 (1972).

[5] J. Sak, *Recursion relations and fixed points for ferromagnets with long-range interactions*, Phys. Rev. B **8**, 281 (1973).

[6] J. M. Kosterlitz, *Phase transitions in long-range ferromagnetic chains*, Phys. Rev. Lett. **37**, 1577 (1976).

[7] E. Luijten and H. Meßingfeld, *Criticality in one dimension with inverse square-law potentials*, Phys. Rev. Lett. **86**, 5305 (2001), arXiv:cond-mat/0104175.

[8] D. Benedetti, E. Lauria, D. Mazáč, and P. van Vliet, *One-dimensional Ising model with $1/r^{1.99}$ interaction*, Phys. Rev. Lett. **134**, 201602 (2025), arXiv:2412.12243.

[9] D. Benedetti, E. Lauria, D. Mazáč, and P. van Vliet, *A strong-weak duality for the 1d long-range Ising model*, SciPost Phys. **20**, 029 (2026), arXiv:2509.05250.

[10] A. J. Leggett, S. Chakravarty, A. T. Dorsey, M. P. A. Fisher, A. Garg, and W. Zwerger, *Dynamics of the dissipative two-state system*, Rev. Mod. Phys. **59**, 1 (1987).

[11] K. G. Wilson, *The renormalization group: Critical phenomena and the Kondo problem*, Rev. Mod. Phys. **47**, 773 (1975).

[12] R. Bulla, N.-H. Tong, and M. Vojta, *Numerical renormalization group for bosonic systems and application to the sub-Ohmic spin-boson model*, Phys. Rev. Lett. **91**, 170601 (2003), arXiv:cond-mat/0306708.

[13] R. Bulla, H.-J. Lee, N.-H. Tong, and M. Vojta, *Numerical renormalization group for quantum impurities in a bosonic bath*, Phys. Rev. B **71**, 045122 (2005), arXiv:cond-mat/0407559.

[14] R. Bulla, T. A. Costi, and T. Pruschke, *Numerical renormalization group method for quantum impurity systems*, Rev. Mod. Phys. **80**, 395 (2008).

[15] M. Vojta, N.-H. Tong, and R. Bulla, *Quantum phase transitions in the sub-Ohmic spin-boson model: Failure of the quantum-classical mapping*, Phys. Rev. Lett. **94**, 070604 (2005); Erratum: Phys. Rev. Lett. **102**, 249904 (2009), arXiv:cond-mat/0410132.

[16] M. Vojta, *Impurity quantum phase transitions*, Philos. Mag. **86**, 1807 (2006).

[17] M. Vojta, R. Bulla, F. Güttge, and F. Anders, *Mass-flow error in the numerical renormalization-group method and the critical behavior of the sub-Ohmic spin-boson model*, Phys. Rev. B **81**, 075122 (2010), arXiv:0911.4490.

[18] M. Vojta, *Numerical renormalization group for the sub-Ohmic spin-boson model: A conspiracy of errors*, Phys. Rev. B **85**, 115113 (2012), arXiv:1201.4922.

[19] C. Guo, A. Weichselbaum, J. von Delft, and M. Vojta, *Critical and strong-coupling phases in one- and two-bath spin-boson models*, Phys. Rev. Lett. **108**, 160401 (2012), arXiv:1110.6314.

[20] A. Winter, H. Rieger, M. Vojta, and R. Bulla, *Quantum phase transition in the sub-Ohmic spin-boson model: Quantum Monte Carlo study with a continuous imaginary time cluster algorithm*, Phys. Rev. Lett. **102**, 030601 (2009), arXiv:0807.4716.

[21] A. Alvermann and H. Fehske, *Sparse polynomial space approach to dissipative quantum systems: Application to the sub-Ohmic spin-boson model*, Phys. Rev. Lett. **102**, 150601 (2009), arXiv:0812.2808.

[22] A. W. Chin, Á. Rivas, S. F. Huelga, and M. B. Plenio, *Exact mapping between system-reservoir quantum models and semi-infinite discrete chains using orthogonal polynomials*, J. Math. Phys. **51**, 092109 (2010).

[23] R. Koekoek, P. A. Lesky, and R. F. Swarttouw, *Hypergeometric Orthogonal Polynomials and Their q-Analogues* (Springer, Berlin, 2010), Sec. 14.12.

[24] K. Fukui and S. Todo, *Order-N cluster Monte Carlo method for spin systems with long-range interactions*, J. Comput. Phys. **228**, 2629 (2009), arXiv:0802.0272.

[25] E. Luijten and H. W. J. Blöte, *Monte Carlo method for spin models with long-range interactions*, Int. J. Mod. Phys. C **6**, 359 (1995).

[26] K. Uzelac, Z. Glumac, and A. Aničić, *Critical behavior of the long-range Ising chain from the largest-cluster probability distribution*, Phys. Rev. E **63**, 037101 (2001), arXiv:cond-mat/0011306.

[27] M. C. Angelini, G. Parisi, and F. Ricci-Tersenghi, *Relations between short-range and long-range Ising models*, Phys. Rev. E **89**, 062120 (2014), arXiv:1401.6805.

[28] V. Pagni, G. Giachetti, A. Trombettoni, and N. Defenu, *One-dimensional long-range Ising model: Two almost equivalent approximations*, Phys. Rev. B **113**, 014406 (2026), arXiv:2510.02458.

A Data formats

Result records (`data/records/<campaign>/<item>.json`; campaign folders `example_K64_B10`, `ds13` (the ds13 campaign), `lambda_pilot`, `lambda_followup`). JSON with three keys:

- `sha256`: SHA-256 of the canonical JSON (sorted keys, no spaces) of the identity and the payload;
- `identity`: configuration (`s`, `Lambda` $= \Lambda$, `nb` $= n_b$, `keep` $= K$, `nsites` $= N$, `tunneling` $= \Omega$, `nlevels`, threads), kind, bracket, fraction (0.3), tolerance, iteration limit, whether the derivative was computed, the solver source hash (`source_sha256`, `3a40deb333e1...`) and the runtime;
- `payload`: `critical_flow` (the flow at α_c with per-site records and the bisection summary), `tuning` (bisection history and final bracket), `tangent` (the derivative flow, with per-site $\partial_\alpha g_n$ and $\ln|\partial_\alpha g_n|$), `analysis` (the window audit: every window with its Δ_E, fit residual and variation, the quality range and the settled window under the key `selected`) and `scope` (a statement of what the audit means).

Per-site records hold the parity gap `odd_gap` $= g_n$, the lowest levels of each sector, the boson occupation and top-level probability of the newest site, and correlations of the spin with the bath. `INDEX.json` in each campaign folder lists the generation (App. B) of every record.

Manifests and receipt (`inputs/manifests`, `inputs/validation_receipt`). A manifest is the frozen list of items submitted to the cluster together with the solver source hash; the receipt is itself a record, issued by the validation job, that binds the source hash to the compute-node runtime. `summarize.py` refuses any record whose identity differs from its manifest item or whose runtime differs from the receipt.

Summaries and results. `make replay` writes per-item summaries (critical coupling, bisection data, quality range, settled window, largest top-level probability and occupation, gate maxima) and per-item details (all windows, per-site gap and $\ln|\partial_\alpha g_n|$) to `.work/data/summaries`. `results/ds13_final.json` holds, per Δ_σ, the estimator, Δ_E, all error components, the settled and extrapolated values with every fit, the Λ values used in δ_Λ, the number of quality windows, α_c, the limiting forms, the superseded midpoint budget, and the late and crossover fits of `crosscheck.py` under `crosschecks`. An extrapolation that is not determined is written as `null`.

B Production on the cluster

All production ran on the AMD nodes of the *symmetry* cluster of the Perimeter Institute (whole nodes with 64 cores and 240 GB, 24 h limit, Slurm batch scheduler). One node with AVX-512 was excluded, since its runtime would not match the receipt. Code was used from a read-only frozen copy with its commit recorded in `GIT_SHA`.

Lanes. One Slurm job (a *lane*) runs many items concurrently on one node, admitting an item while the total threads stay ≤ 64 and the total memory reservation ≤ 232 GiB. Each lane verifies the SHA-256 of its manifest and receipt and logs the start, end, exit code, threads and peak memory of every item (`data/lane_logs`). The memory planner of the code overestimates the need by factors of 1.7 to 3.6 (median 2.8) against the peaks in the lane logs.

Ownership. The shared file system of the cluster does not support file locks across nodes, so items are owned explicitly. Only the submitter, running on the login host under a campaign-wide lock, creates an item's owner file (a random token, a generation number, the job, and the hash of the item definition). A writer must present the current token, and generation g writes only inside its own subdirectory, seeded from the previous generation's checkpoints, so a surviving old process cannot touch new files. An item is reclaimed only when its job is gone from the queue and terminal in the accounting; jobs are never requeued silently.

Validation, smoke runs and drills. The test suite passed on a compute node for each code state, and the last run issued the receipt. A smoke lane of five small items ran packed on one node; a repeat on a second node of the same runtime reproduced it bitwise. At a warning signal 300 s before the wall limit, or by hand, a lane asks every item to checkpoint and exit: an interrupted item resumed from its checkpoint in a second generation and completed; writes without the token were refused from the login node and from another compute node; and two simultaneous reclaim calls produced exactly one winner.

Outage. Two lanes of the Λ follow-up ended with `NODE_FAIL` when every compute node of the cluster stopped responding (3 October 2026, for about 22 h). Their 21 unfinished items were reclaimed as generation 2 and resumed from their checkpoints.

Table 12: Every job of the study (node names anonymized). Thread-h is the sum over items of wall time times threads, from the lane logs; each job held a whole node for its elapsed time.

job	purpose	state	start	elapsed	node	items	thread-h
569091	validation (build)	COMPLETED	2026-10-02 23:34:20	00:01:08	amd-5		
569093	smoke lane	COMPLETED	2026-10-02 23:35:55	00:00:18	amd-5		
569096	USR1 drill, gen. 1	FAILED	2026-10-02 23:36:27	00:00:28	amd-5		
569097	cross-node ownership drill	COMPLETED	2026-10-02 23:36:32	00:00:02	amd-3		
569098	USR1 drill, gen. 2 (resumed)	COMPLETED	2026-10-02 23:37:06	00:03:23	amd-5		
569134	re-validation (fix round 1)	COMPLETED	2026-10-02 23:59:29	00:01:10	amd-6		
569135	second smoke	COMPLETED	2026-10-03 00:01:04	00:00:16	amd-6		
569161	validation of final code	COMPLETED	2026-10-03 00:15:53	00:01:13	amd-6		
569565	example manifest	COMPLETED	2026-10-03 08:48:32	00:08:32	amd-4	3/3	0.6
569572	ds13 lane 0	COMPLETED	2026-10-03 08:54:47	02:39:45	amd-7	27/27	20.4
569573	ds13 lane 1	COMPLETED	2026-10-03 08:54:47	02:22:00	amd-8	29/29	21.7
569574	ds13 lane 2	COMPLETED	2026-10-03 08:54:47	01:34:52	amd-1	30/30	23.1
569575	ds13 lane 3	COMPLETED	2026-10-03 08:54:47	01:09:04	amd-2	30/30	22.7
569576	ds13 lane 4	COMPLETED	2026-10-03 08:54:49	00:58:52	amd-3	30/30	21.8
569602	Λ pilot	COMPLETED	2026-10-03 11:55:38	00:52:39	amd-7	18/18	12.9
569626	Λ follow-up lane 0	NODE_FAIL	2026-10-03 12:52:49	00:15:29	amd-7	3/14	0.2
569627	Λ follow-up lane 1	NODE_FAIL	2026-10-03 12:52:49	00:15:29	amd-8	6/16	0.5
569640	follow-up lane 0, gen. 2	COMPLETED	2026-10-04 11:36:31	02:41:50	amd-2	11/11	27.1
569641	follow-up lane 1, gen. 2	COMPLETED	2026-10-04 11:36:31	02:03:51	amd-3	10/10	26.7

C Provenance and review

Origin of the code. The reference solver and analyzer (`code/lri_spin_boson_hpc/reference/`) and the three archived critical couplings (`code/lri_spin_boson_hpc/inputs/`) come from the original study, an earlier, unreleased study of the project (archive `lri_extended_comparison_study.zip`, SHA-256 `04bb8d30...`; `REFERENCE_PROVENANCE.json`). The optimized package `lri_spin_boson_hpc` was built on it and supplied by the author as the starting point of this study. Both were written by GPT-6 astra (OpenAI), used through ChatGPT; the package itself carries no authorship information.

Review and port to the cluster. Claude Code read the package line by line against the reference solver (no numerical defect; operational defects only) and adapted it to the cluster in a plan–review–build–inspect loop with OpenAI Codex as reviewer. Codex (requested model gpt-6-astra at extra-high reasoning effort; the command-line tool did not report which model ran) approved the plan after four rounds and inspected the built code twice in fresh sessions (10 and 4 findings, all fixed and tested). The last fix round (commit `af9b1dc`: scripts and benchmarks only, solver source unchanged) was not re-inspected. Codex reviewed the cluster port: plan and code. It did not review the numerics, the analysis, the estimator or the table. `REVIEWS.md` summarizes all reviews.

Analysis, checks and this note. Claude Code wrote the analysis scripts (3–8 October), the specification of the second implementation and the comparison scripts (8 October), and drafted this note (13 October; this revision with Claude Opus 5.5). The models of the earlier Claude Code sessions are not recorded in the

package. The second implementation was written by a separate Claude agent from the specification only (Sec. 8.1).

The review of 13 October 2026 and this revision. Three read-only reviews of the previous version of this note (physics, statistics, readability) were made by Claude Opus 5.5 agents, launched one per review from a written brief by a separate Claude Code session that coordinated the public release; they did not see the session that drafted this note or each other's reports. They are independent of each other, not of the model family. They found that the quality-range midpoint sits on the transient before the plateau, that the Λ dependence and the factor 2 were artifacts of it, and that the excess near $s = 1/2$ is slow approach to the fixed point, not NRG truncation. Claude Code verified these findings from the records with its own scripts before changing anything, and the author decided (13 October): (i) the settled late-window value with a chain-depth error, extrapolated where there is no plateau, with the rule fixed before the comparisons were recomputed; (ii) the label "not converged in chain depth" for the three points near $s = 1/2$, which stay shown and unused, the extrapolation at $\Delta_\sigma = 0.225$ being a cross-check only; (iii) the measured Λ spread instead of the factor 2; (iv) the old values in App. E. The use of the midpoint was this study's choice of 4 October, not part of the inherited analyzer, whose own output is the settled window; the package's analyzer calls the quality range "not a physical uncertainty interval".

Second review round (13 October 2026). Three further read-only reviews by the same kind of agents (numbers against the records, note against code on the assembled release, and a recheck of the first round with a scrutiny of the new rule) reproduced every value and error component of Table 1 independently from the raw records, and found the rule's classification robust and the errors not too small. They found that `make replay` failed with the oldest library versions that `requirements.txt` allows (the fits differ in their last bits), several wrong literal numbers and wording issues, and they supplied the fits that are now the late and crossover fits of Sec. 3.8. All were addressed in this version; these last fixes have not been reviewed again.

Reading of 15 October 2026. At the author's request an independent reader without project context, coordinated by a separate Claude Code session, reviewed this note for precision and readability. This version addresses its findings: the status of the label "not converged in chain depth", the definition of the cross-check fits, the one-loop test, the statements about the Monte Carlo comparison and the error budget, the uncertainties near $s = 1$, and notation. These changes were not reviewed again.

Timeline of the choices. The production ran on 3–4 October 2026, after validation on 2 October. On 4 October, after the NRG data had been seen but before any external comparison, Claude Code fixed the first estimate: the quality-range midpoint, the central item, the K/n_b set, the $\Lambda = 2.5$ follow-up, a factor 2 on the Λ shift and a quadrature sum. The comparisons with the ε series and the Monte Carlo study followed on 8 October. The reviews of 13 October showed that the midpoint sits on the transient before the plateau (App. E). The present rule was written the same day, before the comparisons were recomputed, and none of its constants was changed afterwards. It was not written blind, however. The earlier comparisons were known, and the reviews had already compared candidate estimators with the references: the last quality window at $\Delta_\sigma = 0.2$ with the series, an extrapolation at 0.225 with the series, and the plateau value at 0.1 with the Monte Carlo crossing value. The exploration that fixed the rule's constants read only the NRG records. No protocol was registered in advance. The label "not converged in chain depth" was assigned by the author on 13 October, with the series comparison known (Sec. 3.7).

Earlier decision on the points near $s = 1/2$. After the comparisons of 8 October a Claude Code analysis (not released) recommended excluding $\Delta_\sigma = 0.249$ and 0.24 or replacing them by series values, and enlarging the error at 0.225. The author overruled this and kept the three values as computed, with their excess stated, and unused (Sec. 8.2).

The released code tree. `code/lri_spin_boson_hpc` is commit `af9b1dc` (`GIT_SHA`) minus files not needed to test, replay or rerun the study: `adapters/`, the benchmark results and all benchmark scripts except the four that the tests run, `PERFORMANCE.md`, `VALIDATION.md`, `validation/`, `slurm/` (replaced by `NRG/slurm/`), `review/` (summarized in `REVIEWS.md`), `docs/ALGORITHM.md` (Sec. 3 of this note), `cluster/symmetry/freeze.sh` and a duplicate of `inputs/items/example_items.json`. The package's own `README.md` and `docs/SYMMETRY.md` still point to some of these; use `NRG/slurm/` and this note instead. The checksum list `MANIFEST.sha256.json` was regenerated for the remaining files. The solver `src/lri_spin/` is byte-identical to `af9b1dc`, so its SHA-256 still binds every record. Outside the code tree the analysis was revised as described above (`estimator.py` is new; `build_tables.py`, `make_figures.py`, `note_numbers.py`

and `check_figures.py` changed), the comparison scripts take their paths as arguments and no longer compute a quadrature "pull", and C_8 in `compare_methods.py` was updated to the current value, which changes the series references by at most 3×10^{-6} (at $\Delta_\sigma = 0.2$; 2×10^{-7} at 0.225; at $\Delta_\sigma = 0.24$ optimal truncation now stops one order later). `crosscheck.py` (the late and crossover fits) is new and `run_all.sh` now only calls `make replay`.

D All production items

The 194 records of the three campaigns D (ds13, folder `ds13`), P (Λ pilot) and F (Λ follow-up); the three example items are in Table 3. Columns: critical coupling; number of quality windows n_q; quality range; settled value; plateau drift; largest top-level probability p_{edge} over the sites of the critical flow (Sec. 7.2).

campaign	item	α_c	n_q	Δ_E quality range	settled	plateau drift	max p_{edge}
D	ds0.249_L2_K64_B10_N103	0.1071795160	31	[0.51298, 0.51732]	0.513056	3e-04	1e-15
F	ds0.249_L2.5_K64_B10_N78	0.1122397288	12	[0.50803, 0.51056]	0.508166	5e-04	2e-13
F	ds0.249_L2.5_K128_B10_N78	0.1122368788	13	[0.50808, 0.51105]	0.508078	6e-04	2e-13
F	ds0.249_L2.5_K128_B16_N78	0.1122377084	12	[0.50789, 0.51046]	0.508151	6e-04	2e-21
D	ds0.249_L3_K64_B8_N65	0.1173295097	13	[0.51730, 0.51890]	0.517479	3e-04	2e-03
D	ds0.249_L3_K48_B10_N65	0.1173263397	5	[0.50910, 0.50996]	0.509096	6e-04	5e-12
D	ds0.249_L3_K64_B10_N65	0.1173260725	4	[0.50892, 0.50966]	0.508918	7e-04	9e-04
D	ds0.249_L3_K96_B10_N65	0.1173255595	5	[0.50906, 0.50995]	0.509058	6e-04	5e-12
D	ds0.249_L3_K128_B10_N65	0.1173254942	4	[0.50886, 0.50965]	0.509122	8e-04	9e-04
D	ds0.249_L3_K64_B12_N65	0.1173260502	3	[0.50814, 0.50862]	0.508138	8e-04	1e-04
D	ds0.249_L3_K64_B16_N65	0.1173261829	2	[0.50800, 0.50827]	0.508000	8e-04	2e-04
D	ds0.249_L3_K128_B16_N65	0.1173255478	2	[0.50803, 0.50828]	0.508030	8e-04	4e-19
D	ds0.249_L3_K64_B20_N65	0.1173261902	3	[0.50815, 0.50860]	0.508155	7e-04	6e-24
D	ds0.249_L3_K64_B24_N65	0.1173261918	2	[0.50798, 0.50826]	0.507985	9e-04	1e-28
D	ds0.249_L4_K64_B10_N52	0.1269362680	0	–	0.510640	1e-03	5e-03
D	ds0.24_L2_K64_B10_N103	0.1166435376	15	[0.50223, 0.50699]	0.503006	8e-04	7e-06
F	ds0.24_L2.5_K64_B10_N78	0.1222308934	13	[0.49911, 0.50160]	0.499379	6e-04	2e-13
F	ds0.24_L2.5_K128_B10_N78	0.1222282882	13	[0.49902, 0.50166]	0.499125	5e-04	2e-13
F	ds0.24_L2.5_K128_B16_N78	0.1222289164	13	[0.49819, 0.50145]	0.498993	8e-04	2e-05
D	ds0.24_L3_K64_B8_N65	0.1278472683	13	[0.50788, 0.50929]	0.507908	2e-04	1e-09
D	ds0.24_L3_K48_B10_N65	0.1278438327	5	[0.49962, 0.50057]	0.499625	7e-04	6e-12
D	ds0.24_L3_K64_B10_N65	0.1278435265	5	[0.49960, 0.50056]	0.499602	7e-04	6e-12
D	ds0.24_L3_K96_B10_N65	0.1278430054	5	[0.49982, 0.50058]	0.499944	6e-04	6e-12
D	ds0.24_L3_K128_B10_N65	0.1278429538	5	[0.49964, 0.50057]	0.499643	7e-04	6e-12
D	ds0.24_L3_K64_B12_N65	0.1278435098	4	[0.49906, 0.49967]	0.499222	7e-04	2e-14
D	ds0.24_L3_K64_B16_N65	0.1278436618	4	[0.49913, 0.49967]	0.499434	8e-04	3e-19
D	ds0.24_L3_K128_B16_N65	0.1278429945	4	[0.49892, 0.49962]	0.498922	7e-04	3e-19
D	ds0.24_L3_K64_B20_N65	0.1278436705	2	[0.49899, 0.49929]	0.498992	9e-04	9e-05
D	ds0.24_L3_K64_B24_N65	0.1278436707	0	–	0.499254	9e-04	6e-05
D	ds0.24_L4_K64_B10_N52	0.1384395657	0	–	0.501245	9e-04	5e-04
D	ds0.225_L2_K64_B10_N103	0.1339530997	30	[0.49078, 0.49557]	0.491797	4e-04	7e-06
F	ds0.225_L2.5_K64_B10_N78	0.1405232485	17	[0.49033, 0.49270]	0.490435	3e-04	2e-13
F	ds0.225_L2.5_K128_B10_N78	0.1405208812	17	[0.48974, 0.49231]	0.489830	3e-04	2e-13
F	ds0.225_L2.5_K128_B16_N78	0.1405214369	17	[0.48974, 0.49220]	0.489897	4e-04	7e-22
D	ds0.225_L3_K64_B8_N65	0.1471196556	14	[0.49618, 0.49761]	0.496210	2e-04	1e-09
D	ds0.225_L3_K48_B10_N65	0.1471158185	9	[0.48991, 0.49132]	0.489911	4e-04	8e-04
D	ds0.225_L3_K64_B10_N65	0.1471154781	9	[0.48988, 0.49131]	0.489880	4e-04	8e-04
D	ds0.225_L3_K96_B10_N65	0.1471149602	9	[0.48979, 0.49134]	0.490139	5e-04	8e-04
D	ds0.225_L3_K128_B10_N65	0.1471149168	9	[0.48973, 0.49132]	0.490102	5e-04	8e-04
D	ds0.225_L3_K64_B12_N65	0.1471154761	9	[0.48927, 0.49099]	0.489689	6e-04	5e-04
D	ds0.225_L3_K64_B16_N65	0.1471156481	9	[0.48931, 0.49098]	0.489697	6e-04	2e-04
D	ds0.225_L3_K128_B16_N65	0.1471149209	9	[0.48924, 0.49098]	0.489668	6e-04	2e-04
D	ds0.225_L3_K64_B20_N65	0.1471156611	8	[0.48944, 0.49098]	0.489856	7e-04	1e-04
D	ds0.225_L3_K64_B24_N65	0.1471156769	9	[0.48956, 0.49099]	0.489559	4e-04	7e-30
D	ds0.225_L4_K64_B10_N52	0.1595443361	3	[0.49095, 0.49129]	0.490952	6e-04	3e-10
P	ds0.2_L1.5_K64_B10_N176	0.1626225790	89	[0.53623, 0.53696]	0.536303	1e-07	2e-14
P	ds0.2_L1.5_K128_B10_N176	0.1603705735	76	[0.50652, 0.50855]	0.507049	1e-05	3e-10
P	ds0.2_L1.5_K64_B16_N176	0.1626210299	89	[0.53623, 0.53696]	0.536303	1e-07	3e-26
D	ds0.2_L2_K64_B10_N103	0.1676281822	34	[0.48357, 0.48710]	0.484345	2e-04	5e-06
P	ds0.2_L2_K128_B10_N103	0.1676098637	36	[0.48306, 0.48637]	0.483192	9e-05	3e-14
F	ds0.2_L2_K192_B10_N103	0.1676082707	34	[0.48256, 0.48641]	0.483363	2e-04	5e-06
F	ds0.2_L2_K256_B10_N103	0.1676081346	35	[0.48261, 0.48637]	0.483290	2e-04	5e-06
P	ds0.2_L2.5_K64_B10_N78	0.1761692489	23	[0.48295, 0.48558]	0.483530	2e-04	1e-04
P	ds0.2_L2.5_K128_B10_N78	0.1761670918	22	[0.48338, 0.48541]	0.483420	1e-04	2e-13
F	ds0.2_L2.5_K128_B16_N78	0.1761675436	23	[0.48307, 0.48538]	0.483233	1e-04	2e-22
D	ds0.2_L3_K64_B8_N65	0.1847272377	17	[0.48643, 0.48778]	0.486523	1e-04	1e-03
D	ds0.2_L3_K48_B10_N65	0.1847227575	15	[0.48333, 0.48467]	0.483334	1e-04	6e-04
D	ds0.2_L3_K64_B10_N65	0.1847222031	15	[0.48341, 0.48467]	0.483442	1e-04	5e-12
D	ds0.2_L3_K96_B10_N65	0.1847218040	15	[0.48350, 0.48468]	0.483517	2e-04	5e-12
D	ds0.2_L3_K128_B10_N65	0.1847217711	15	[0.48344, 0.48467]	0.483457	1e-04	5e-12
D	ds0.2_L3_K64_B12_N65	0.1847223271	15	[0.48332, 0.48457]	0.483330	1e-04	1e-14
D	ds0.2_L3_K64_B16_N65	0.1847224093	15	[0.48330, 0.48457]	0.483321	1e-04	7e-20
D	ds0.2_L3_K128_B16_N65	0.1847217485	15	[0.48288, 0.48458]	0.483392	2e-04	1e-04
D	ds0.2_L3_K64_B20_N65	0.1847225105	15	[0.48292, 0.48457]	0.483316	2e-04	7e-05
D	ds0.2_L3_K64_B24_N65	0.1847225110	15	[0.48302, 0.48457]	0.483326	2e-04	3e-08
D	ds0.2_L4_K64_B10_N52	0.2008115424	8	[0.48366, 0.48435]	0.483664	2e-04	3e-03
D	ds0.175_L2_K64_B10_N103	0.2084469699	40	[0.48393, 0.48650]	0.484362	4e-05	3e-06
F	ds0.175_L2.5_K64_B10_N78	0.2194635288	27	[0.48411, 0.48571]	0.484116	4e-05	2e-13
F	ds0.175_L2.5_K128_B10_N78	0.2194614482	25	[0.48362, 0.48532]	0.483990	6e-05	9e-05

campaign	item	α_c	n_q	Δ_E quality range	settled	plateau drift	max p_{edge}
D	ds0.175_L3_K64_B8_N65	0.2304807533	19	[0.48562, 0.48647]	0.485623	3e-05	6e-05
D	ds0.175_L3_K48_B10_N65	0.2304755377	19	[0.48403, 0.48505]	0.484042	6e-05	5e-12
D	ds0.175_L3_K64_B10_N65	0.2304749169	19	[0.48395, 0.48505]	0.483987	4e-05	5e-04
D	ds0.175_L3_K96_B10_N65	0.2304744981	19	[0.48403, 0.48506]	0.484044	7e-05	5e-12
D	ds0.175_L3_K128_B10_N65	0.2304744735	19	[0.48400, 0.48505]	0.484007	4e-05	5e-12
D	ds0.175_L3_K64_B12_N65	0.2304749974	19	[0.48395, 0.48501]	0.483951	3e-05	1e-14
D	ds0.175_L3_K64_B16_N65	0.2304751345	19	[0.48396, 0.48501]	0.483974	5e-05	2e-20
D	ds0.175_L3_K128_B16_N65	0.2304744393	19	[0.48377, 0.48502]	0.483986	8e-05	1e-04
D	ds0.175_L4_K64_B10_N52	0.2511400536	11	[0.48414, 0.48461]	0.484150	9e-05	2e-10
D	ds0.15_L2_K64_B10_N103	0.2580502776	42	[0.49114, 0.49310]	0.491365	2e-05	2e-06
F	ds0.15_L2.5_K64_B10_N78	0.2721823788	29	[0.49118, 0.49233]	0.491181	8e-06	8e-13
F	ds0.15_L2.5_K128_B10_N78	0.2721802333	28	[0.49111, 0.49226]	0.491113	2e-05	8e-13
D	ds0.15_L3_K64_B8_N65	0.2862878455	21	[0.49191, 0.49271]	0.492051	3e-05	7e-04
D	ds0.15_L3_K48_B10_N65	0.2862812698	21	[0.49102, 0.49187]	0.491127	3e-05	2e-04
D	ds0.15_L3_K64_B10_N65	0.2862807211	21	[0.49112, 0.49187]	0.491127	2e-05	4e-12
D	ds0.15_L3_K96_B10_N65	0.2862803155	21	[0.49105, 0.49187]	0.491120	2e-05	3e-04
D	ds0.15_L3_K128_B10_N65	0.2862802913	21	[0.49099, 0.49187]	0.491123	3e-05	3e-04
D	ds0.15_L3_K64_B12_N65	0.2862808093	21	[0.49110, 0.49185]	0.491099	1e-05	7e-15
D	ds0.15_L3_K64_B16_N65	0.2862809076	21	[0.49105, 0.49185]	0.491099	2e-05	2e-08
D	ds0.15_L3_K128_B16_N65	0.2862802558	21	[0.49110, 0.49185]	0.491106	2e-05	5e-06
D	ds0.15_L4_K64_B10_N52	0.3126750056	14	[0.49118, 0.49169]	0.491181	4e-05	2e-10
D	ds0.125_L2_K64_B10_N103	0.3186931853	45	[0.50588, 0.50735]	0.505878	4e-06	2e-12
F	ds0.125_L2.5_K64_B10_N78	0.3367673600	31	[0.50570, 0.50675]	0.505714	6e-06	1e-09
F	ds0.125_L2.5_K128_B10_N78	0.3367650662	31	[0.50567, 0.50672]	0.505676	3e-06	3e-08
D	ds0.125_L3_K64_B8_N65	0.3547738461	23	[0.50623, 0.50693]	0.506258	1e-05	2e-04
D	ds0.125_L3_K48_B10_N65	0.3547649005	23	[0.50569, 0.50640]	0.505687	8e-06	6e-12
D	ds0.125_L3_K64_B10_N65	0.3547642364	23	[0.50568, 0.50640]	0.505687	1e-05	6e-12
D	ds0.125_L3_K96_B10_N65	0.3547638772	23	[0.50568, 0.50640]	0.505685	6e-06	6e-12
D	ds0.125_L3_K128_B10_N65	0.3547638514	23	[0.50568, 0.50640]	0.505683	5e-06	6e-12
D	ds0.125_L3_K64_B12_N65	0.3547643489	23	[0.50567, 0.50639]	0.505675	6e-06	5e-15
D	ds0.125_L3_K64_B16_N65	0.3547644468	23	[0.50567, 0.50639]	0.505676	6e-06	1e-11
D	ds0.125_L3_K128_B16_N65	0.3547638114	23	[0.50567, 0.50639]	0.505672	9e-07	5e-21
D	ds0.125_L4_K64_B10_N52	0.3883742704	15	[0.50571, 0.50609]	0.505710	1e-05	2e-10
P	ds0.1_L1.5_K64_B10_N176	0.3786792319	87	[0.56695, 0.56733]	0.567059	6e-08	6e-12
P	ds0.1_L1.5_K128_B10_N176	0.3733064368	83	[0.53919, 0.54017]	0.539196	4e-07	6e-12
P	ds0.1_L1.5_K64_B16_N176	0.3786839607	87	[0.56695, 0.56733]	0.567059	7e-08	5e-22
D	ds0.1_L2_K64_B10_N103	0.3936704020	46	[0.52980, 0.53105]	0.529796	2e-06	9e-12
P	ds0.1_L2_K128_B10_N103	0.3936438035	46	[0.52962, 0.53089]	0.529631	2e-06	9e-12
F	ds0.1_L2_K192_B10_N103	0.3936429089	46	[0.52961, 0.53087]	0.529613	3e-06	4e-11
F	ds0.1_L2_K256_B10_N103	0.3936426886	46	[0.52960, 0.53087]	0.529609	2e-06	9e-12
P	ds0.1_L2.5_K64_B10_N78	0.4167902745	32	[0.52964, 0.53056]	0.529637	1e-06	7e-11
P	ds0.1_L2.5_K128_B10_N78	0.4167874486	32	[0.52961, 0.53054]	0.529610	2e-06	4e-11
D	ds0.1_L3_K64_B8_N65	0.4397848811	24	[0.53000, 0.53063]	0.529998	2e-06	2e-06
D	ds0.1_L3_K48_B10_N65	0.4397704546	23	[0.52962, 0.53006]	0.529616	3e-06	2e-11
D	ds0.1_L3_K64_B10_N65	0.4397697893	23	[0.52961, 0.53006]	0.529613	1e-06	2e-08
D	ds0.1_L3_K96_B10_N65	0.4397694732	23	[0.52961, 0.53006]	0.529613	3e-07	2e-11
D	ds0.1_L3_K128_B10_N65	0.4397694449	23	[0.52961, 0.53006]	0.529612	2e-06	3e-09
D	ds0.1_L3_K64_B12_N65	0.4397699011	23	[0.52961, 0.53006]	0.529609	2e-06	6e-11
D	ds0.1_L3_K64_B16_N65	0.4397699761	23	[0.52961, 0.53006]	0.529611	5e-06	2e-14
D	ds0.1_L3_K128_B16_N65	0.4397693853	23	[0.52961, 0.53006]	0.529608	2e-06	4e-21
D	ds0.1_L4_K64_B10_N52	0.4825812310	16	[0.52962, 0.52999]	0.529628	6e-06	2e-10
D	ds0.075_L2_K64_B10_N111	0.4882361312	51	[0.56644, 0.56743]	0.566440	6e-07	4e-11
F	ds0.075_L2.5_K64_B10_N84	0.5179599027	36	[0.56628, 0.56707]	0.566284	5e-07	7e-11
F	ds0.075_L2.5_K128_B10_N84	0.5179561214	36	[0.56625, 0.56705]	0.566260	7e-07	7e-11
D	ds0.075_L3_K64_B8_N70	0.5474812521	28	[0.56655, 0.56708]	0.566546	4e-07	6e-08
D	ds0.075_L3_K48_B10_N70	0.5474526311	28	[0.56627, 0.56681]	0.566265	6e-07	1e-10
D	ds0.075_L3_K64_B10_N70	0.5474520316	28	[0.56624, 0.56681]	0.566265	2e-06	1e-10
D	ds0.075_L3_K96_B10_N70	0.5474517839	28	[0.56626, 0.56681]	0.566263	9e-07	1e-10
D	ds0.075_L3_K128_B10_N70	0.5474517479	28	[0.56625, 0.56681]	0.566262	1e-06	1e-10
D	ds0.075_L3_K64_B12_N70	0.5474520799	28	[0.56626, 0.56681]	0.566261	1e-06	1e-13
D	ds0.075_L3_K64_B16_N70	0.5474521815	28	[0.56626, 0.56681]	0.566262	1e-06	4e-17
D	ds0.075_L3_K128_B16_N70	0.5474516143	28	[0.56625, 0.56681]	0.566260	1e-06	6e-20
D	ds0.075_L4_K64_B10_N55	0.6022718795	18	[0.56627, 0.56657]	0.566270	6e-07	7e-08
P	ds0.05_L1.5_K64_B10_N217	0.5909590331	100	[0.67067, 0.67118]	0.670692	3e-07	1e-10
P	ds0.05_L1.5_K128_B10_N217	0.5780543351	102	[0.63035, 0.63087]	0.630353	1e-07	1e-10
P	ds0.05_L1.5_K64_B16_N217	0.5909603112	100	[0.67066, 0.67119]	0.670693	3e-07	9e-20
D	ds0.05_L2_K64_B10_N127	0.6120267877	59	[0.62230, 0.62295]	0.622300	5e-08	2e-10
P	ds0.05_L2_K128_B10_N127	0.6119924970	59	[0.62205, 0.62277]	0.622125	4e-07	2e-10
F	ds0.05_L2_K192_B10_N127	0.6119916485	59	[0.62211, 0.62277]	0.622119	8e-08	2e-10
F	ds0.05_L2_K256_B10_N127	0.6119911289	59	[0.62208, 0.62277]	0.622116	3e-07	2e-10
P	ds0.05_L2.5_K64_B10_N96	0.6507599139	43	[0.62214, 0.62266]	0.622139	1e-07	3e-10
P	ds0.05_L2.5_K128_B10_N96	0.6507542509	43	[0.62206, 0.62264]	0.622116	4e-07	3e-10
D	ds0.05_L3_K64_B8_N80	0.6892086918	34	[0.62234, 0.62282]	0.622349	2e-07	2e-07
D	ds0.05_L3_K48_B10_N80	0.6891363592	34	[0.62211, 0.62260]	0.622120	3e-07	6e-10
D	ds0.05_L3_K64_B10_N80	0.6891357316	34	[0.62212, 0.62260]	0.622119	3e-07	6e-10
D	ds0.05_L3_K96_B10_N80	0.6891355001	34	[0.62208, 0.62260]	0.622117	7e-07	6e-10
D	ds0.05_L3_K128_B10_N80	0.6891354544	34	[0.62212, 0.62260]	0.622117	2e-07	6e-10
D	ds0.05_L3_K160_B10_N80	0.6891354015	34	[0.62210, 0.62260]	0.622117	5e-07	6e-10
D	ds0.05_L3_K192_B10_N80	0.6891353149	34	[0.62212, 0.62260]	0.622117	2e-07	1e-09
D	ds0.05_L3_K64_B12_N80	0.6891353868	34	[0.62212, 0.62260]	0.622117	3e-07	4e-12
D	ds0.05_L3_K64_B16_N80	0.6891355208	34	[0.62212, 0.62260]	0.622117	4e-07	9e-19
D	ds0.05_L3_K128_B16_N80	0.6891349793	34	[0.62210, 0.62259]	0.622115	5e-07	9e-19
D	ds0.05_L4_K64_B10_N63	0.7603342269	23	[0.62211, 0.62236]	0.622122	1e-06	1e-09
D	ds0.025_L2_K64_B10_N143	0.7879305686	63	[0.71355, 0.71403]	0.713586	6e-07	1e-09
F	ds0.025_L2.5_K64_B10_N108	0.8401456498	47	[0.71340, 0.71387]	0.713448	7e-07	2e-09
F	ds0.025_L2.5_K128_B10_N108	0.8401346083	47	[0.71338, 0.71384]	0.713422	7e-07	2e-09
D	ds0.025_L3_K64_B8_N90	0.8921068317	38	[0.71361, 0.71402]	0.713648	7e-07	9e-07
D	ds0.025_L3_K48_B10_N90	0.8918504547	38	[0.71339, 0.71380]	0.713425	7e-07	3e-09

campaign	item	α_c	n_q	Δ_E quality range	settled	plateau drift	max p_{edge}
D	ds0.025_L3_K64_B10_N90	0.8918498061	38	[0.71339, 0.71380]	0.713424	7e-07	3e-09
D	ds0.025_L3_K96_B10_N90	0.8918496024	38	[0.71339, 0.71380]	0.713423	7e-07	3e-09
D	ds0.025_L3_K128_B10_N90	0.8918495416	38	[0.71339, 0.71380]	0.713423	7e-07	3e-09
D	ds0.025_L3_K160_B10_N90	0.8918494283	38	[0.71339, 0.71380]	0.713423	7e-07	3e-09
D	ds0.025_L3_K192_B10_N90	0.8918492517	38	[0.71338, 0.71380]	0.713423	7e-07	3e-09
D	ds0.025_L3_K64_B12_N90	0.8918472563	38	[0.71339, 0.71380]	0.713422	7e-07	9e-12
D	ds0.025_L3_K64_B16_N90	0.8918474343	38	[0.71339, 0.71380]	0.713423	7e-07	2e-17
D	ds0.025_L3_K128_B16_N90	0.8918469295	38	[0.71339, 0.71380]	0.713422	7e-07	2e-17
D	ds0.025_L4_K64_B10_N71	0.9876957393	26	[0.71341, 0.71366]	0.713427	1e-06	8e-09
D	ds0.01_L2_K64_B10_N166	0.9503649771	66	[0.80960, 0.81027]	0.809951	8e-06	5e-09
F	ds0.01_L2.5_K64_B10_N126	1.0158010470	49	[0.80949, 0.81011]	0.809784	7e-06	8e-09
F	ds0.01_L2.5_K128_B10_N126	1.0157777118	49	[0.80945, 0.81008]	0.809749	7e-06	8e-09
D	ds0.01_L3_K64_B8_N105	1.0814250819	41	[0.80974, 0.81041]	0.810031	1e-05	3e-06
D	ds0.01_L3_K48_B10_N105	1.0806604206	41	[0.80946, 0.81013]	0.809751	9e-06	1e-08
D	ds0.01_L3_K64_B10_N105	1.0806597653	41	[0.80946, 0.81013]	0.809749	9e-06	1e-08
D	ds0.01_L3_K96_B10_N105	1.0806595638	41	[0.80946, 0.81013]	0.809749	9e-06	1e-08
D	ds0.01_L3_K128_B10_N105	1.0806594892	41	[0.80946, 0.81013]	0.809749	9e-06	1e-08
D	ds0.01_L3_K160_B10_N105	1.0806592593	41	[0.80946, 0.81013]	0.809749	9e-06	1e-08
D	ds0.01_L3_K192_B10_N105	1.0806588400	41	[0.80946, 0.81013]	0.809748	9e-06	1e-08
D	ds0.01_L3_K64_B12_N105	1.0806490638	41	[0.80946, 0.81013]	0.809748	9e-06	5e-11
D	ds0.01_L3_K64_B16_N105	1.0806491182	41	[0.80946, 0.81013]	0.809748	9e-06	2e-16
D	ds0.01_L3_K128_B16_N105	1.0806486418	41	[0.80946, 0.81012]	0.809747	9e-06	2e-16
D	ds0.01_L4_K64_B10_N83	1.2010212236	30	[0.80947, 0.81002]	0.809756	1e-05	3e-08
D	ds0.001_L2_K64_B10_N396	1.1270646401	150	[0.93780, 0.93799]	0.937991	4e-06	2e-08
F	ds0.001_L2.5_K64_B10_N300	1.2079818456	125	[0.93729, 0.93752]	0.937523	3e-06	3e-08
F	ds0.001_L2.5_K128_B10_N300	1.2078925099	126	[0.93719, 0.93742]	0.937418	3e-06	3e-08
D	ds0.001_L3_K64_B8_N250	1.2911723978	105	[0.93793, 0.93813]	0.938132	2e-06	7e-06
D	ds0.001_L3_K48_B10_N250	1.2885532639	106	[0.93722, 0.93743]	0.937429	2e-06	5e-08
D	ds0.001_L3_K64_B10_N250	1.2885527136	106	[0.93722, 0.93743]	0.937428	2e-06	5e-08
D	ds0.001_L3_K96_B10_N250	1.2885525078	106	[0.93722, 0.93743]	0.937428	2e-06	5e-08
D	ds0.001_L3_K128_B10_N250	1.2885524382	106	[0.93722, 0.93743]	0.937428	2e-06	5e-08
D	ds0.001_L3_K160_B10_N250	1.2885515952	106	[0.93722, 0.93743]	0.937427	2e-06	5e-08
D	ds0.001_L3_K192_B10_N250	1.2885497372	106	[0.93722, 0.93743]	0.937425	2e-06	5e-08
D	ds0.001_L3_K64_B12_N250	1.2885056066	106	[0.93722, 0.93742]	0.937424	2e-06	3e-10
D	ds0.001_L3_K64_B16_N250	1.2885051164	106	[0.93722, 0.93742]	0.937424	2e-06	2e-15
D	ds0.001_L3_K128_B16_N250	1.2885046794	106	[0.93722, 0.93742]	0.937424	2e-06	2e-15
D	ds0.001_L4_K64_B10_N198	1.4387239566	78	[0.93724, 0.93751]	0.937441	8e-06	1e-07

E The superseded midpoint estimator

Until 13 October 2026 the value of a point was the midpoint of the quality range of its central item, with the half-width δ_{w} as one error component and the total $\delta_{\text{tot}}^{\text{old}} = [\delta_{\text{w}}^2 + \delta_K^2 + (2\delta_\Lambda^{\text{old}})^2]^{1/2}$, where $\delta_\Lambda^{\text{old}}$ was the midpoint shift from $\Lambda = 3$ to 2.5. Claude Code chose it when the final table was first built (4 October); the inherited analyzer reports the settled window. Table 14 lists the old values.

Why it was replaced. Along the quality windows Δ_E falls monotonically at 11 of 13 points (Fig. 7): the early quality windows are the transient toward the fixed point. The midpoint therefore lies 1.7 to 3.7×10^{-4} above the plateau at $0.025 \le \Delta_\sigma \le 0.15$ (and -1.0×10^{-4} from it at $\Delta_\sigma = 0.001$, where the windows rise), and δ_{w} measures the length of the transient, not a resolution of the plateau; the earlier claim of a precision floor of about 10^{-4} was wrong. Because the window rule counts sites while the chain length scales as $1/\ln\Lambda$, a smaller Λ admits a larger share of transient windows, which produced an apparent rise of Δ_E as Λ decreases (the midpoint at $\Lambda = 2$ lay $+4.0 \times 10^{-4}$ and $+6.5 \times 10^{-5}$ above $\Lambda = 3$ at $\Delta_\sigma = 0.1$ and 0.05, against -3×10^{-6} and -1×10^{-6} on plateau values). The factor 2 on $\delta_\Lambda^{\text{old}}$ multiplied this artifact. At $0.01 \le \Delta_\sigma \le 0.15$ the new errors are smaller than the old ones by factors 6–16. At 13 of the 13 points the old value lies within its old $\delta_{\text{tot}}^{\text{old}}$ of the new one, but only because δ_{w} equalled the offset: the old interval was in effect one-sided.

F Terms and symbols

Abbreviations. LRI: long-range Ising; NRG: numerical renormalization group; RG: renormalization group; MC: Monte Carlo; BLAS: basic linear algebra subprograms; SHA-256: a cryptographic checksum.

term	meaning
point	one value of Δ_σ
item, record	one NRG calculation at fixed $(\Delta_\sigma, \Lambda, K, n_b, N)$; the JSON file holding its result
central item	the item at $\Lambda = 3$, $K = 128$, $n_b = 16$
campaign, production	a set of items submitted together; the released items
finite map	the truncated NRG at fixed Λ, K, n_b, N as a function of α
window, quality window	w consecutive sites over which $\ln\lvert\partial_\alpha g_n\rvert$ is fitted; a window with relative variation < 0.005 and fit residual < 0.003
quality range	the range of Δ_E over the quality windows of an item

term	meaning
settled window, settled value	the window with the smallest fit residual plus relative variation; its Δ_E
plateau, plateau drift	a point whose central item has plateau drift $\leq \tau_{\mathrm{p}}$; the largest change of Δ_E within $\pm h$ sites of the settled window
extrapolated	a point without a plateau; its value is the extrapolation of Δ_{loc} (Sec. 3.7)
not converged in chain depth	label assigned by the author to $\Delta_\sigma = 0.249$, 0.24, 0.225, not an outcome of the estimator: the flow does not reach its fixed point within $l_{\max}$; no error quoted, not used
trusted window	a window whose finite-difference Δ_E in the second implementation is stable under a change of the step η
late fit, crossover fit	cross-check fits of Δ_{loc} with a detuning term (Sec. 3.8); not adopted
n.d.	not determined: an extrapolation whose fitted ω is below 0.01
fit residual, terminal residual, eigen-residual	largest deviation from the window fit; $f(\alpha_c)$; $\max \lvert HU - UE \rvert$
top-level probability p_{edge}	weight of the highest level $n_b - 1$ of the newest oscillator in the ground state, at one site of the critical flow
receipt, manifest, identity, runtime	see App. A
lane, generation, drill	see App. B

symbol	meaning
d, s	dimension (1); exponent of the coupling decay $1/r^{1+s}$ and of the bath spectrum $\rho \propto \omega^s$
Δ_σ, ε, δ	spin dimension $(1-s)/2$; $2s - 1 = 1 - 4\Delta_\sigma$; $1 - s = 2\Delta_\sigma$
Δ_ε, y_t, ν, ω	dimension of the energy operator; thermal RG eigenvalue $1/\nu$; $\Delta_\varepsilon = 1 - y_t$; leading correction exponent (also, in $\rho(\omega)$ and ω_j, a frequency)
$C_n^{\mathrm{SR}}(1)$	coefficients of the dual expansion (4) of Δ_ε in $\sqrt{\Delta_\sigma}$
u, l_0, $\mathcal{N}$	φ^4 coupling of the one-loop flow; RG time where it is infinite; number of components of the O($\mathcal{N}$) model
Δ_E, $\Delta_E^{\mathrm{central}}$	NRG estimate of Δ_ε; that of the central item
Δ_{loc}, l, $l_{\max}$	local exponent, Eq. (14); RG time $n \ln \Lambda$; its usable maximum, Eq. (15)
Δ_∞, A, l_0, l_{hi}, S	extrapolated value, amplitude, first and last RG time and span of an extrapolation fit
A_{d}, l_{a}, l_{end}, k, n_{fit}	detuning amplitude, first and last RG time, end offset k/y_t and number of accepted fits of a cross-check fit
τ_{p}, h	plateau threshold; half-width of the plateau neighborhood, $\lceil w/4 \rceil$ sites
τ	imaginary time (Sec. 2)
C_n	coefficients of the ε expansion of Δ_ε for $d = 1$
Ω, α, α_c, $\delta\alpha$	tunneling amplitude (0.1); dissipation strength; its critical value for the finite map; residual detuning
$\rho(\omega)$, $J(\omega)$, ω_j, λ_j, b_j	bath spectral density; $\pi\rho$; oscillator frequencies, couplings and annihilation operators
Λ, q, I_m	discretization ratio; $1/\Lambda$; frequency bins
c_m, γ_m, ξ_m, ξ_0	star oscillators, their couplings and frequencies $\xi_m = \xi_0 q^m$
e_n, t_n, $\bar{e}_n$, $\bar{t}_n$	chain on-site energies and hoppings; rescaled by Λ^n
$\mathcal{A}_n$, $\mathcal{B}_n$, a	little q-Jacobi recurrence quantities of Eq. (8); $a = q^s$
λ_{tot}	total coupling of the spin to the bath, $\sqrt{2\alpha/(s+1)}$
H_n, $\bar{b}_n$, $\mathbb{1}$	rescaled Hamiltonian of sites $0, \ldots, n$; site-n operator in the kept basis; identity
K, n_b, N, $N(\Lambda)$	kept states per parity sector; oscillator levels; chain sites; chain sites at Λ
P	parity $\sigma_x (-1)^{\sum b^\dagger b}$
g_n, $E_{0,n}^{\mathrm{odd/even}}$	parity gap; lowest level of H_n in each sector
$f(\alpha)$	bisection function, Eq. (11)
U_k, U_d, E_k, E_d	kept and discarded eigenvectors and energies
$\dot{H}$, Z, T, $\dot{E}_0$, B	$\partial_\alpha H$; rotation of the kept states; derivative of the kept Hamiltonian; of the ground energy; site operator
w, κ, n_q	window width; intercept of the window fit; number of quality windows
δ_{depth}, δ_K, δ_Λ, δ_{tot}	chain-depth error; K/n_b spread; discretization error; their sum

term	meaning
δ_{w}, $\delta_{\Lambda}^{\mathrm{old}}$, $\delta_{\mathrm{tot}}^{\mathrm{old}}$	quantities of the superseded estimator (App. E)
$n_{\max}$	estimated last site near the fixed point in double precision
σ_{stat}, σ_{sys}, σ_{∞}, $\Delta_{\omega'}$	statistical error and systematic scale of the Monte Carlo fit; error and sensitivity of its crossing extrapolation (not a dimension)
η	relative step of the finite differences of the second implementation (`h` in its code)
L, M, u	Padé orders and Borel variable
c1–c8; D, P, F	configurations of the second implementation; campaign tags of App. D

Table 14: The superseded estimator. Old value: quality-range midpoint with $\delta_{\mathrm{tot}}^{\mathrm{old}}$ (two significant digits, as given in the earlier version of this note); the half-width δ_{w} and $2|\delta_{\Lambda}^{\mathrm{old}}|$ in units of 10^{-4}; the value of this note; the difference; the old status labels: (a) far above the ε series compared with the old error, attributed to a truncation bias of bosonic NRG, not used; (b) above the Borel–Padé sum by about the old error, same attribution, not used; (c) at $\Delta_\sigma = 0.2$, the series cannot test the point at the 10^{-3} level. The attribution to a truncation bias has since been withdrawn (Sec. 8.2).

Δ_σ	old: midpoint ($\delta_{\mathrm{tot}}^{\mathrm{old}}$)	$10^4\delta_{\mathrm{w}}$	$10^4 \cdot 2\delta_{\Lambda}^{\mathrm{old}}$	this note	$10^4\times$(old − this note)	old status
0.249	0.5082(21)	1.3	20.5	[0.508030]	+1.3	biased (a)
0.24	0.4993(12)	3.5	10.9	[0.498922]	+3.5	biased (a)
0.225	0.4901(19)	8.7	17.2	[0.489668]	+4.4	biased (b)
0.2	0.4837(13)	8.5	9.9	0.4830(7)	+7.5	final (c)
0.175	0.48440(65)	6.2	1.1	0.48390(35)	+4.9	final
0.15	0.49148(64)	3.8	5.1	0.49111(11)	+3.7	final
0.125	0.50603(48)	3.6	3.2	0.50567(5)	+3.6	final
0.1	0.52983(52)	2.3	4.7	0.529608(33)	+2.2	final
0.075	0.56653(36)	2.8	2.4	0.566260(30)	+2.7	final
0.05	0.62235(25)	2.5	0.2	0.622115(27)	+2.3	final
0.025	0.71359(21)	2.1	0.3	0.713422(29)	+1.7	final
0.01	0.80979(34)	3.3	0.6	0.80975(5)	+0.4	final
0.001	0.93732(11)	1.0	0.4	0.93742(11)	-1.0	final